QIDDIYA DEVELOPMENT COMPANY

DOCUMENT NUMBER:	[illegible]	REVISION NUMBER:	00	ISSUED BY:	PMO
DOCUMENT TITLE:	MIR FORM	DOCUMENT TYPE:	FORM	DATE ISSUED:	MAY-08-2022

Qiddiya

MATERIAL INSPECTION REQUEST (MIR) FORM

Contractor:	HAIF COMPANY
Project:	QPMO-410-CT-00618- ROAD ACCESS TO RESORT CORE
Contract No.:	QI09043311-WO-0004
Material Inspection Request Number:	QRXM114000-HFC-TEC-MIR-ITS-02-RO5-00007

Haif Company | Freyssinet | HAIF-FREYSSINET JV

Discipline

Architecture	☐	Mechanical	☐	Information Technology	☐	Quality	☐	Project Controls	☐
Civil/Structural	☑	Electrical	☐	Landscape	☐	Safety	☐	Construction Management	☐
Other	☐								

MATERIAL DESCRIPTION

Inspection / Test Location:	INTERIM ROAD MAIN LINE-02		
Material Name:	ITS SYSTEMS LOWERING DEVICE,	Quantity:	03 Pcs.
P.O Number		Delivery Note Number:	TTS-HAIF-DN0013
Approved Material Transmittal Number: (Aconex document Approval)	QRXM114000-HFC-TEC-MAT-ITS-02-RO5-00005	Approval Date:	
Inspection Date:	02-Apr-2026	Inspection Start Time:	09:00 AM
ITP:		Checklist:	
Spec's / Dwg. Ref.:	TRAFFIC TECH SECURITY W.L.L	Approved Method Statement Number:	
Attachments:	Test Certificate: ◉ Yes ○ No IFC / Shop Drawings: ○ Yes ◉ No		

DESCRIPTION OF MATERIAL

Lowering Device for 15 Mtrs pole 28,7042 CDP ARM6" TEM 15M (03 Pcs.) Interim Road Main Line-02

Three pieces.

CONTRACTOR - QUALITY INSPECTOR - INITIAL INSPECTION FINDINGS:

Are Complete Details Mentioned in P.O and Delivery Note?	◉ YES ○ NO	REMARKS	
Complies with Approved ITPs/Checklists?	◉ YES ○ NO	REMARKS	
Signs of Physcial Damage?	○ YES ◉ NO	REMARKS	
Conforms to Approved Material Submittal?	○ YES ○ NO	REMARKS	

Accepted:	◉
Accepted with Comments:	○
Rejected (See Remarks):	○

REMARKS:					
Name:	Moataz Elkhateeb	Signature:		Date:	02-04-2026

QIDDIYA / SUPERVISION CONSULTANT QC INSPECTOR - FINAL INSPECTION

B: Approved with Comments
QI09043311-UPR TRN17&TNS05
Saud Nawab
Zuhair Fayez Partnership Consultants
Riyadh , Saudi Arabia

Contractor Has complied with Material requirements?	○ YES ○ NO	ctly?	✓ YES ○ NO ○ NA		
Accepted:					○
Accepted with Comments:					✓
Rejected (See Remarks):					○
REMARKS:	① Camera lowering device with accessories should be compatible with CCTV cameras. ② Operating tools shall be handed over to QIC operation & maintenance team.				
Name:	M. YASIR.	Signature:		Date	02/04/2026.

FOR DETAILED MATERIAL INSPECTION STATUS, PLEASE SEE ATTACHED CHECKLIST.
CHECKLISTS THAT ARE NOT ATTACHED AND SIGNED BY SUPERVISION CONSULTANT WILL VOID THIS INSPECTION REQUEST.

This document is the property of Qiddiya Development Company, 2022

Client:	Consultant:	Contractor:
Qiddiya	Zuhair Fayez Partnership	Haif Company / Freyssinet — HAIF-FREYSSINET JV

TECHNICAL DOCUMENTS

Project Name: Package 3 - **Interim Road Access to Resort Core**	**Project Code:** QRXM114000
Location: 00	**Date:** 17-September-2025
Document Title: **Material Approval Request FOR ITS SYSTEM SUPPLYING & INSTALLING BY TRAFFIC TECH SECURITY W.L.L Traffic Surveillance CCTV**	**Ref No:** QRXM114000-HFC-TEC-MAT-ITS-02-RO5-00005
	Rev No: 00

Department: ☐ Architectural ☐ Structural ☐ Electrical ☐ Mechanical ☑ ITS ☐ Civil

Type: ☐Pre-Qualification ☐ Method Statement ☒ Material Submittal ☐ Sample ☐ Other

Document Description: Material Approval Request FOR ITS SYSTEM SUPPLYING & INSTALLING BY TRAFFIC TECH SECURITY W.L.L Traffic Surveillance CCTV

Doc. Ref. No.	Title
QRXM114000-HFC-TEC-MAT-ITS-02-RO5-00005	**Material Approval Request FOR ITS SYSTEM SUPPLYING & INSTALLING BY TRAFFIC TECH SECURITY W.L.L Traffic Surveillance CCTV**

Enclosure:

Submitted By:	**Received By:**
Name: **Eng. Moataz**	Name:
Signature:	Signature:
Date: 17-Sep-2025	Date:

Below portion for Consultant Supervision use only

Status: ☐A- Approved ☐B- Approved as noted ☐C- Revise & Resubmit ☐R- For information only

Comments:

B: Approved with Comments — Zuhair Fayez Partnership Consultants Riyadh, Saudi Arabia

It is a conditional approval, as changes are expected in the design, specifications, and drawings. The Contractor/Supplier must ensure proper coordination and compliance with expected changes & following comments prior to procurement. In case of any contradiction or non-compliance, this approval shall be deemed Code-C, and the Contractor/Supplier shall rectify it without any additional cost or time impact.

1. Provide traffic surveillance system storage calculation in accordance to project specification & attach data sheet for storage equipment.
2. The traffic surveillance system shall send event, alert, and alarm notifications to the VSS.
3. Provide reference that the subject material has Category C1 as mentioned in the submittal.
4. Camera shall have internal storage to save video of at least 24hr with the 30FPS in-case of network failure
5. Attach Riser Diagram of Traffic surveillance system.
6. Attach SASO/SABER Certificate of proposed material
7. Provide Warranty certificate for five years from the satisfactory completion of the experimental period as per project specification

B: Approved with Comments — Saud Nawab — Zuhair Fayez Partnership Consultants Riyadh, Saudi Arabia

Below portion for Client use only

Comments:

B: Approved with Comments — Zuhair Fayez Partnership Consultants Riyadh, Saudi Arabia

Signed by: Signature: Date:

TRAFFIC TECH
www.traffic-tech.com

تقنيات المرور الأمنية ذ.م.م.
TRAFFIC TECH SECURITY W.L.L.

Date: 05-01-2026
Our Ref: TTS-HAIF_DN-0013

Delivery Note

To: HAIF COMPANY
---------- الى:

Ref: WS045CR-53 (RA2RC – ACCESS ROAD TO RESORT CORE)

SN	Model No. رقم الموديل	Description الوصـــــف	Unit الوحـدة	Qty الكميـة	Remarks ملاحظـات
MATERIAL SUPPLY (ITS SYSTEMS)					
1	CDP6-6A15M	LOWERING DEVICE FOR 15 MTRS POLE 28,7042 CDP ARM 6" TEM 15M	NOS	03	

Delivered By: EHAB BASLA EHAB ---------- **Date:** 05 January 2026 ----------
Printed Name & Signature

Received By: ----------
Printed Name & Signature

Date: ----------

Hail Contracting
Date
Project. Qiddiya
WS-045CR
Received
Qty check

شركة تقنيات المرور الأمنية ذ.م.م
TRAFFIC TECH
سجل تجاري ١٠١٠٨٨٢٧٣٢
TRAFFIC TECH Security W.L.L. Co.

NORTH STAR LIGHTING, LLC,
DEPARTMENT NO. 10219 PO 87618, CHICAGO , ILLINOIS
60680-0618 USA

PACKING LIST

SUPPLIER / SHIPPER :	*NORTH STAR LIGHTING, LLC,* *DEPARTMENT NO, 10219 PO 87618, CHICAGO , ILLINOIS* *60680-0618 USA.*	NO. OF ORDER:	POR-18353
		NO. OF SHIPMENT :	1
CONSIGNEE / BUYER :	**M/S. TRAFFIC TECH (GULF) W.L.L.** **P.O.BOX 30704 DOHA QATAR**	DELIVERY TO /PORT OF DISCHARGE:	DOHA INTERNATIONAL AIRPORT, DOHA, QATAR
		DATE :	15-Dec-2025
DESCRIPTION OF GOODS :	DOHA INTERNATIONAL AIRPORT, DOHA, QATAR CAMERA LOWERING SYSTEM AS PER PROFORMA INVOICE 137256		

CONTAINER NO.	TYPE OF STRUCTURE	NO. OF ITEMS	NO. OF PACKAGES (WOODEN CRATES)	DIMENSION OF WOODEN CRATES LxWxH (cm.)	TOTAL NET WEIGHT (KG)	TOTAL GROSS WEIGHT (KG)
	CDP6 CAMERA LOWERING SYSTEMS	5	1	127X93.98X101.6	144	225
SEAL NO.						
	TOTAL :	**5**	**1**		**144**	**225**

NO OF PACKAGES :1 pcs	NET WEIGHT 144 kg.	GROSS WEIGHT 225 kg.

NORTH STAR LIGHTING LLC CORPORATE SEAL 2012 DELAWARE

DONALD L PIKE
DIRECTOR OF SALES
NORTH STAR LIGHTING, LLC

December 15,2025

CAMERA
LOWERING
SYSTEMS

TEST REPORTS
&
INSTALLATION
MANUAL

Camera Lowering Systems
835 Industrial Drive • Elmhurst, IL 60126 USA • PH: 708-681-4330 • Fax 708-681-4006 • www.Lowering-Device.com

CAMERA LOWERING SYSTEMS

Installation Manual	
Part No.	Description
CDP Series	Pole Mounting Systems

Revised By	Number	DATE
DONALD PIKE	F	08/22/23

INSTALLATION AND OPERATING MANUAL FOR:

CDP SERIES
POLE MOUNTING SYSTEMS

- POLE AND ARM FITTER 1
- COMPOSITE CABLE HANDLING 2
- CDP INSTALLATION INSTRUCTIONS 3-4
- CABLE CLAMP INSTRUCTIONS 5-6
- CDP INSTALLATION INSTRUCTIONS 6-8
- ARM LEVELING INSTRUCTIONS 9
- WIRE ROPE HANDLING INSTRUCTIONS 10
- DRILL AND CLUTCH SPECIFICATIONS 11-12
- PREVENTATIVE MAINTENANCE 13
- OPERATING INSTRUCTIONS 14-15

NOTE: - The installer must provide the 1.5" PVC conduit, PVC solvent cement and PVC fittings.
-The conduit must be attached to the conduit fitting provided by North Star Lighting.
-The raising and lowering cable must run inside the conduit.
-The conduit must terminate approximately 1 foot above the hand hole.

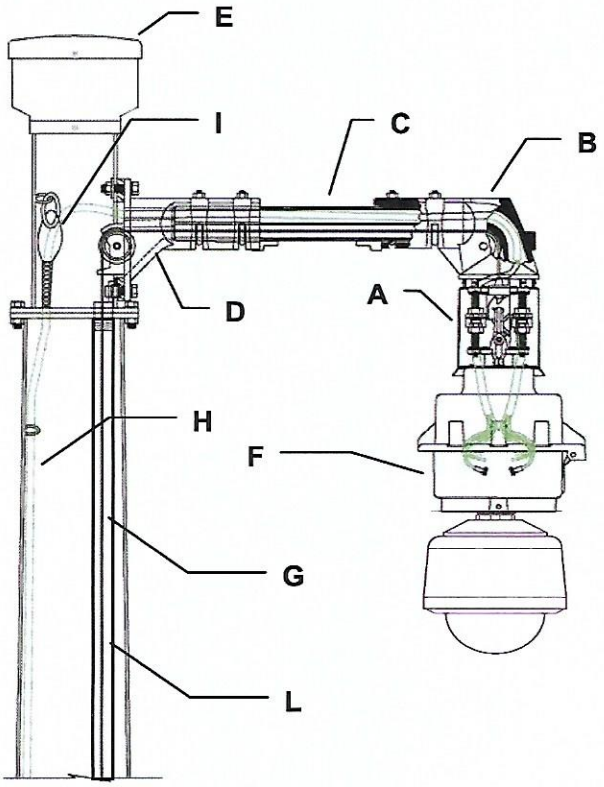
SAFETY WARNING
Please follow all state and local safety regulations while you are performing this work.
Reading and understanding this manual will significantly reduce the risk of injury.

NORTH STAR LIGHTING, LLC ◆ PH: 708-681-4330 ◆ Fax 708-681-4006 ◆ www.lowering-device.com

CAMERA LOWERING SYSTEMS

Design CDP6-16HDDN50-85SC

Arm and Disconnect Unit for Pole Mounting Multi-Function Surveillance Cameras



A. ELECTRICAL AND SIGNAL DISCONNECT UNI WITH MULTI-CONTACT CONNECTOR.

B. DISCONNECT UNIT FITTER WITH PULLEY AND U-BOLT MOUNTING FOR INTERLOCKING NON-ROTATING ARM.

C. DIVIDED PIPE ARM: 2-3/8 IN. O.D. SEPARATES CONTROL CABLE AND ELECTRICAL AND SIGNAL WIRES. POSITION ALIGNED NON-ROTATING.

D. POLE MOUNTED FITTER WITH PULLEY AND U-BOLT MOUNTING. FITTER MOUNTS DIRECTLY TO 6 INCH O.D. TENON AT TOP OF POLE.

E. POLE TOP JUNCTION BOX WITH COVER SLIPS OVER 6" O.D. TENON.

F. CAMERA CONNECTION BOX PROVIDED WITH STABILIZING WEIGHTS. EASY OPEN SWING DOWN DESIGN PERMITS QUICK ACCESS TO ELECTRICAL/SIGNAL WIRES FROM CAMERA ASSEMBLY.

G. CONTROL CABLE CONSTRUCTED OF 5/32 INCH DIA. STAINLESS STEEL 7 X 19 CONSTRUCTION CABLE INSIDE 1 ½" CONDUIT (CONDUIT BY OTHERS).

H. ONE PIECE SIGNAL CORD WITH COAX AND POWER FROM DISCONNECT UNIT TO POLE BASE OR TO CABINET WITHOUT THE NEED OF EXTRA CONNECTORS.

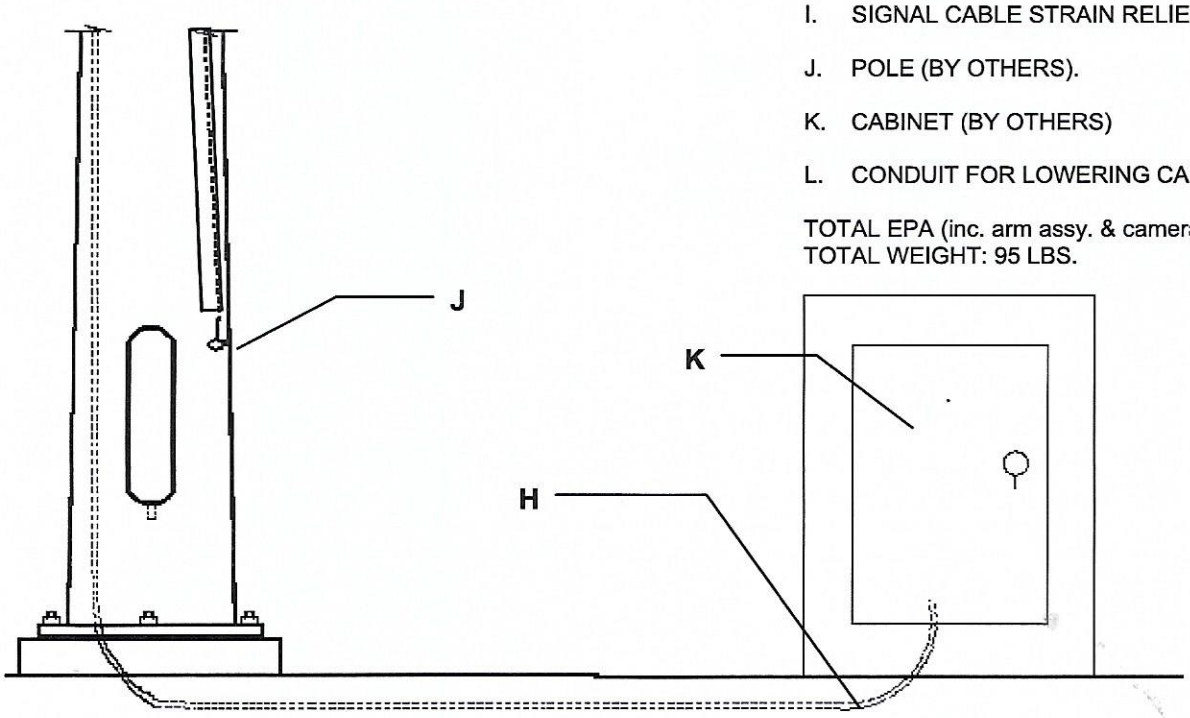
I. SIGNAL CABLE STRAIN RELIEF

J. POLE (BY OTHERS).

K. CABINET (BY OTHERS)

L. CONDUIT FOR LOWERING CABLE (BY OTHERS)

TOTAL EPA (inc. arm assy. & camera): 2.50
TOTAL WEIGHT: 95 LBS.



CAMERA LOWERING SYSTEMS

COMPOSITE CABLE HANDLING RIGHT and WRONG

The composite cable assembly consists of the coaxial cable, electrical power cables, twisted pair, and/or ethernet signal cable.

The composite cable is very delicate because of the signal carrying properties and should be handled with care.

There is a correct way to handle the composite cable assy and a wrong way.

DO NOT carry the CDP assembly by the cable. Pressure can be place on the cable, causing damage, possible breakage of wires. Also, the cable might pull off or partially off the crimped contact connector.

ALWAYS carry the CDP assembly as shown on the left. Take the pressure off the cable by putting the roll over the shoulder.

The CDP should be carried with both hands carrying the arm assembly as shown.

CAMERA LOWERING SYSTEMS

CDP INSTALLATION INSTRUCTIONS

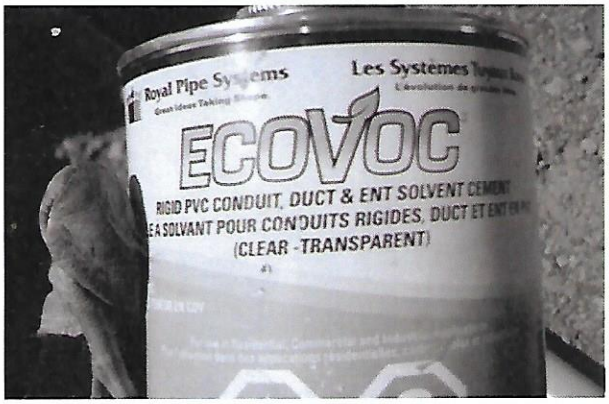
PREPARATION

1. Remove all items from cartons.
2. Make sure all component parts have been shipped. The CLS system consists of:
   A. The arm/disconnect/cable assembly
   B. Camera Junction Box
   C. Pole Top Junction Box (if ordered)
   D. Lowering tool assembly
   E. Drill & Clutch (if ordered)
3. Have tools needed for installation.
4. Have 1 1/2" PVC and glue ready provided by Installer .

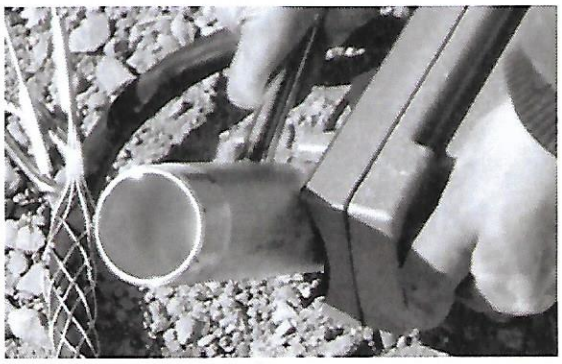
5. Pole must be positioned in a horizontal position, close to the installation location. It is preferred to have the top of the pole resting on a construction horse.
6. Concrete and Anchor bolts should have been poured.
7. Crane should be ready.

ASSEMBLY INSTRUCTIONS

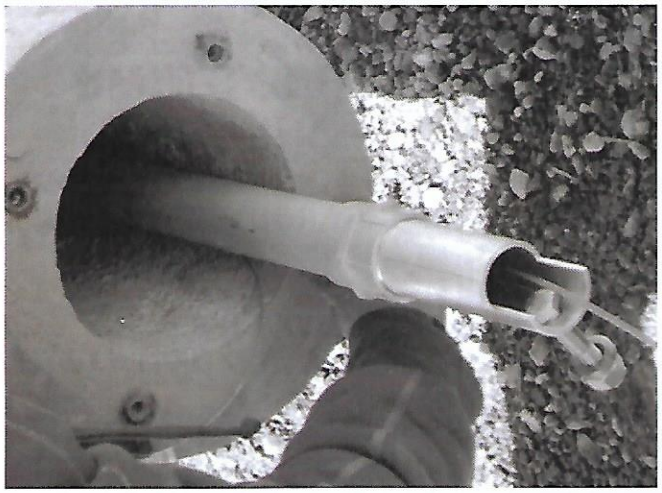
1. First, glue the PVC tubes together and allow glue to dry.



2. Use a 'Rigid PVC Solvent Cement' to secure the PVC tubes together.
3. Install PVC into the pole from the top.
4. Trim off access PVC.



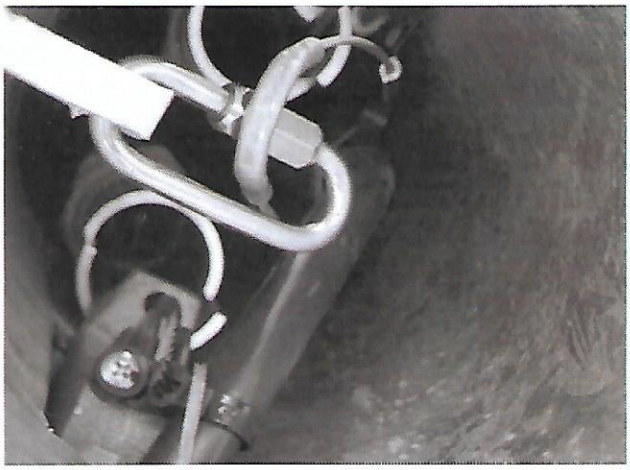
5. Remove PVC adaptor from 'Arm Assembly'.



6.Connect conduit to conduit adaptor.
7.Unroll raising and lowering stainless steel cable like a garden hose. Careful not to kink the cable.

8.Using a 'fish tape', fish the cable thru the arm opening in the tenon, and into the conduit.
9.Attach the end of the cable with the connecting link, to the eye-bolt parking stand close to the handhole.
10.Unwind the composite (electrical/signal) cable like a garden hose.
11.With the help of the fish tape, feed the composite cable thru the arm opening in the tenon, and down the inside of the pole. Use eye-bolt guides to keep the cable close to the wall of the pole.

12.Bolt the arm to the tenon with the two 1/2" bolts provided.
13.Bolt the conduit adaptor to the lower 1/2" bolt.



14.Connect the strain relief to the eyebolt in the tenon. Adjust the strain relief so as to allow slack in the electrical cable feeding into the arm.

15.Bolt tenon to top of pole. Make sure the arm is not underneath the handholes near the base of the pole.
16.With the raising and lowering cable attached to the eyebolt near the base of the pole, loosen the cable clamp screws in the disconnect unit.
17.Pull the raising and lowering cable thru the clamp until there is only a small amount of slack (approx. 2-3").
18.Reassemble the cable onto the cable clamp and tighten the screws (see attached drawing)

CAMERA LOWERING SYSTEMS

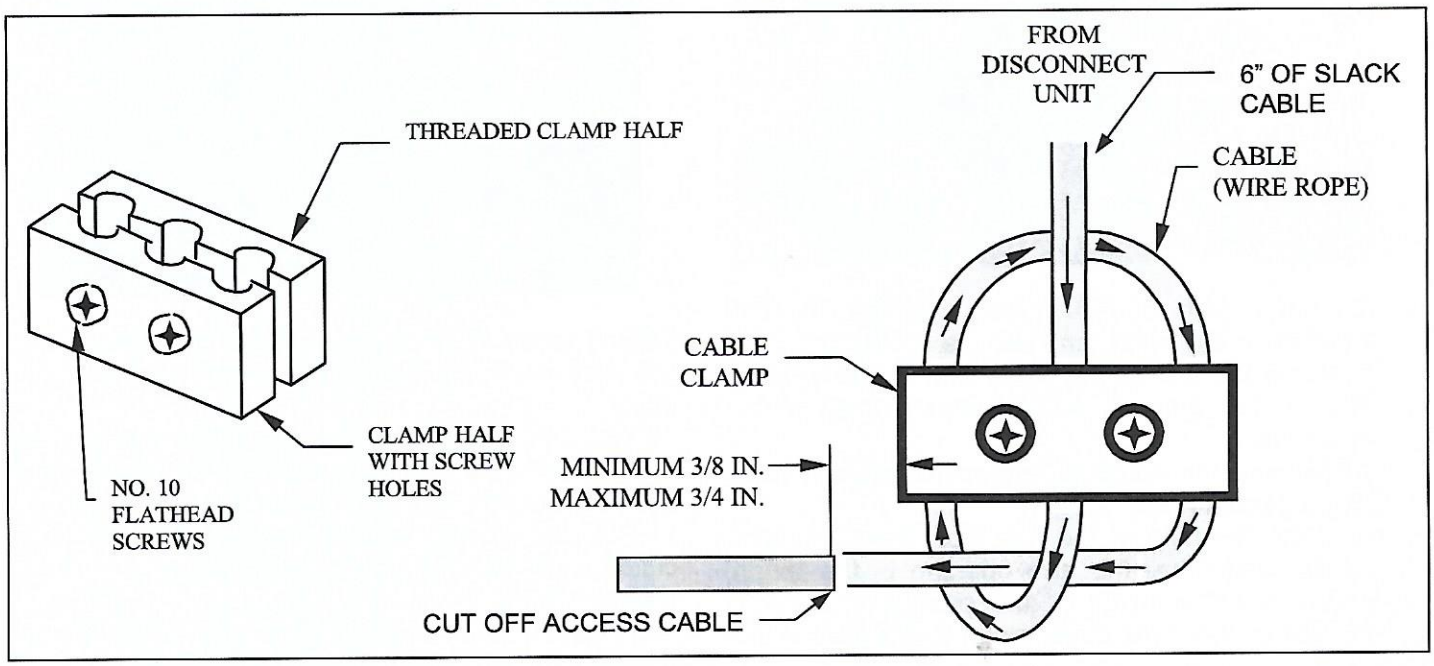
INSTALLATION AND OPERATING MANUAL:

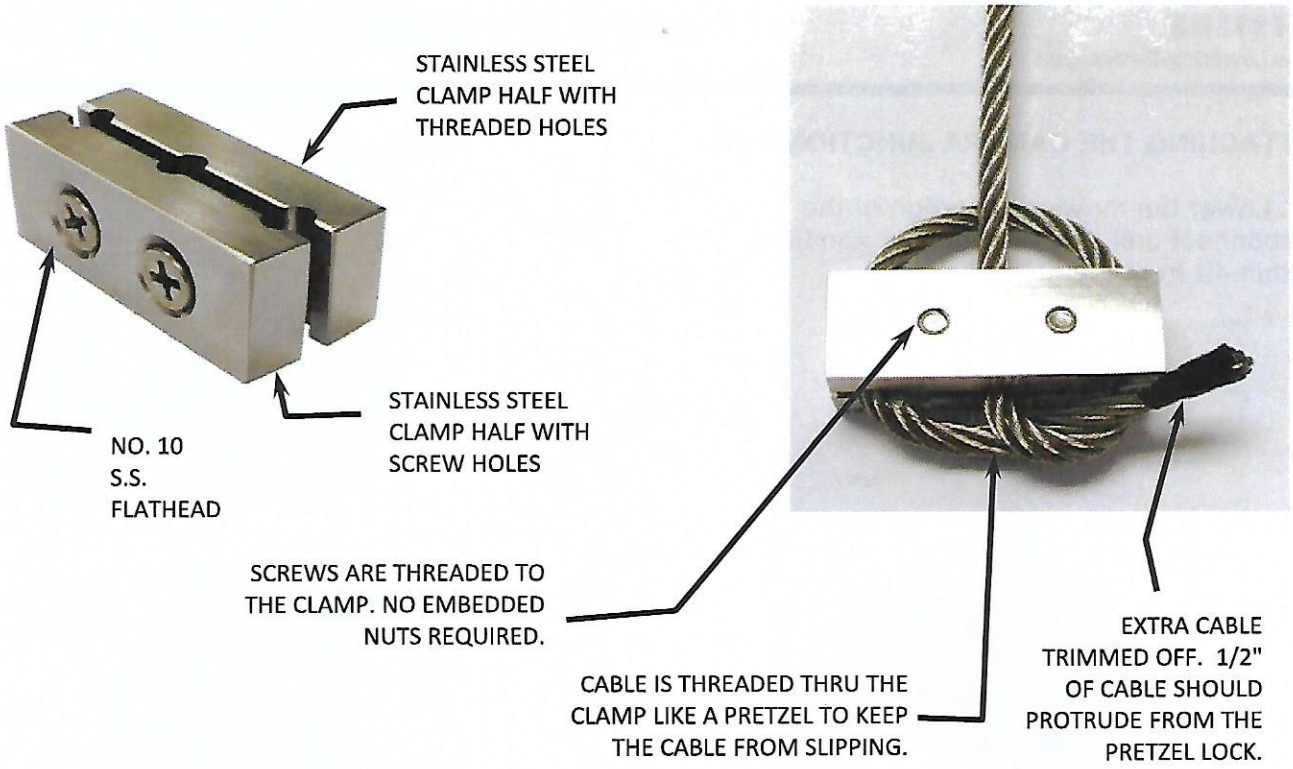
CABLE CLAMP ASSEMBLY AND USE FOR SYSTEMS WITH LIFT CAPACITIES UP TO 400 POUNDS

WARNING: ● **TO PREVENT SERIOUS PERSONAL INJURY, THE CABLE CLAMP MUST BE PROPERLY ASSEMBLED AND THE CABLE MUST BE PROPERLY ROUTED THROUGH THE CLAMP AS SPECIFIED WITHIN. NO ATTEMPT AT LIFTING ANY LOAD SHOULD BE MADE UNTIL ALL SPECIFICATION CONDITIONS ARE MET.**

- **DO NOT LIFT PEOPLE OR OPERATE ANY LOWERING SYSTEM OVER PEOPLE.**
- **USE ONLY 1/8 INCH OR 5/32 INCH DIAMETER 7x19 CONSTRUCTION CABLE WITH THIS CLAMP.**
- **THE ASSEMBLY WILL COME WITH CABLE LONGER THAN IS NEEDED.**
- **BEFORE THE FIXTURE IS ATTACHED, ATTACH THE QUICK LINK ON THE OTHER END OF THE CABLE TO THE EYEBOLT PARKING STAND NEAR THE HANDHOLE.**
- **THE CABLE SHOULD ONLY HAVE A SMALL AMOUNT OF SLACK (APPROX 4") WHEN THE DISCONNECT UNIT IS IN THE LOCKED (RAISED) POSITION.**
- **LOOSEN THE two No.10 SCREWS AND PULL THE CABLE THRU AS SHOWN BELOW.**
- **WHEN YOU HAVE 3" OF CABLE SLACK, TIGHTEN UP THE TWO No. 10 SCREWS.**
- **PUT ELECTRICAL TAPE AROUND THE CABLE AND CUT OFF ACCESS CABLE FROM THE END OF THE CABLE CLAMP. THE TAPE WILL KEEP CABLE FROM FRAYING.**

FIXING CABLE IN CLAMP



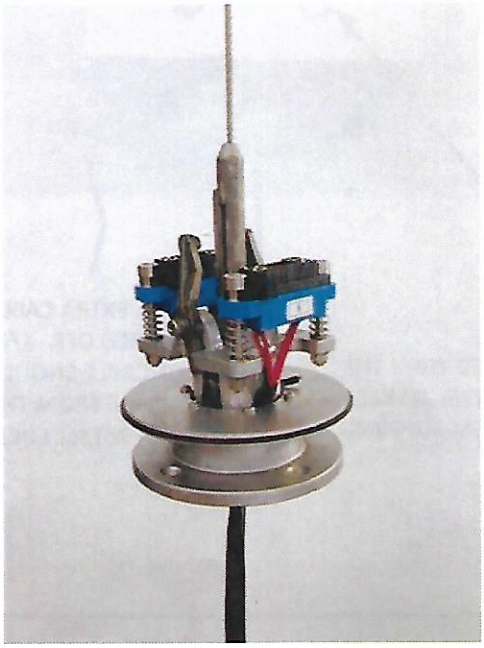


19. Cut the raising and lowering cable.
20. Reassemble the disconnect unit.
21. Put the disconnect unit in the locked position and tie a cord to the bolts on the bottom of the dis- connect unit.
22. Mount the Pole Top Junction Box on top of the tenon.
23. Now the pole should be ready to be hoisted up.
24. After the pole is up and the anchor bolts are se- cured to the pole (or pole secured to ground in di- rect burial poles), bolt the lowering tool to the pole utilizing the 1/2" bolt.
25. Unwind several feet of cable from the lowering tool.
26. Connect the lowering tool cable to the camera lowering cable.
27. Unhook the cable from the pole eyebolt.
28. With the handle, manually lower the moveable portion of the disconnect unit by first cranking to remove the slack and raising the system an additional 3/4".
29. Reverse direction and lower the unit.
30. Gently pull on the cord attached to the move- able portion to unlock the unit, while lowering the unit.

ATTACHING THE CAMERA JUNCTION BOX

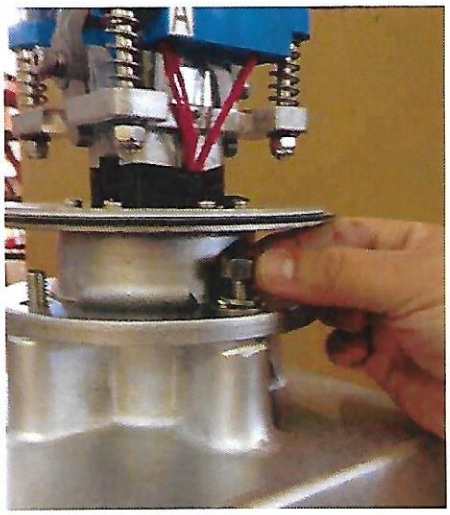
Chapter 5

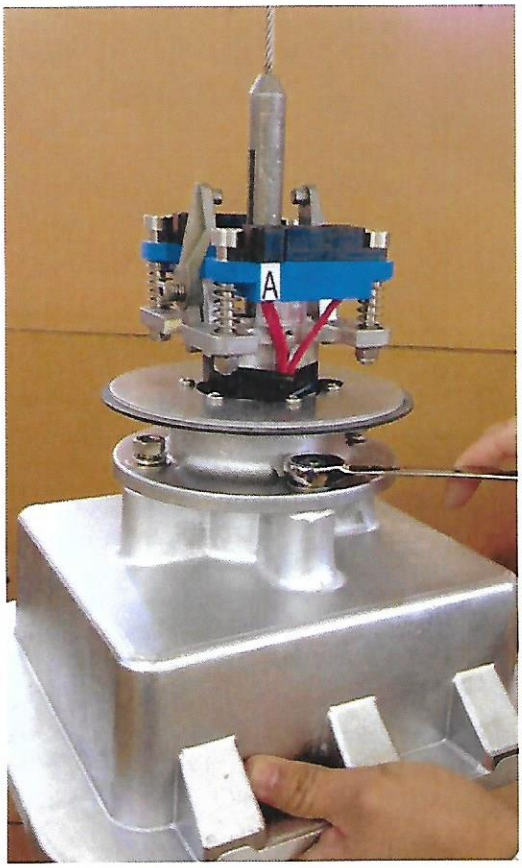
01. Lower the moveable portion of the disconnect unit down to working condition within 4ft from the ground.



02. Unhinge the two clam shells of the Camera Junction box from each other.

03. Put the communication cable from the lower half of the disconnect unit thru the opening of the upper half of the Camera Junction Box.
04. Align the four 3/8" threaded rods of the Camera Junction Box with the mounting flange of the lowered disconnect unit.
05. Attach washers and nuts provided, onto the threaded rod. After two nuts have been partially threaded onto the rods, it is not necessary to hold the upper half of the junction box.
06. Attach the rest of the hardware onto the rods and tighten.
07. Attach the bottom half of the junction box to the upper half with the 2 hinges.
08. Insert the communication wires into the box, and other hardware.



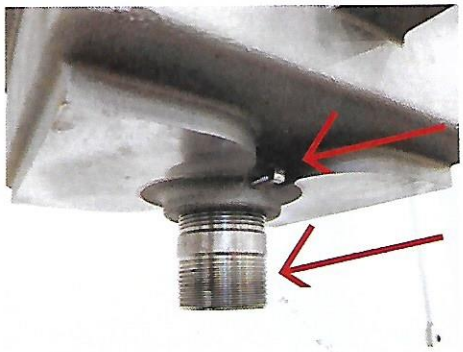


09. Close the box utilizing the stainless steel lock by turning the lock until it stops.
10. The box is now attached.
11. Check the center guide pin and make sure that it is well lubricated. It should come from the factory lubricated.

12. The Camera Junction Box has weights inside the box to allow the system to be lowered.
13. Raise the unit to the top. The drill can be used at this time. The drill should not be used during the operation of the last 1-2 feet. Manually raise the unit to the top.
14. Then reverse direction and lower the unit until there is slack in the cable.
15. Repeat the lowering and raising operations several times to make sure the system is operating properly.
16. If the camera is available, connect the camera to the junction box when lowered, and connect the signal/power cable.

MOUNTING THE CAMERA

17. When using a 1 ½" threaded stem or the FLA-110 flange to mount the camera to the bottom of the Camera Junction Box, make sure to thread the male stem as far as possible.
18. Then USE **LOCTITE RED 271** on the 2 set screws provided, to tighten the stem to the CAMJB-10, to eliminate vibration that could loosen the stem or flange.



LOCTITE Threadlocker RED 271

ADDITIONAL NOTES

Make sure that the pole and the arm assembly is level. The system will not work if the disconnect unit is not level.

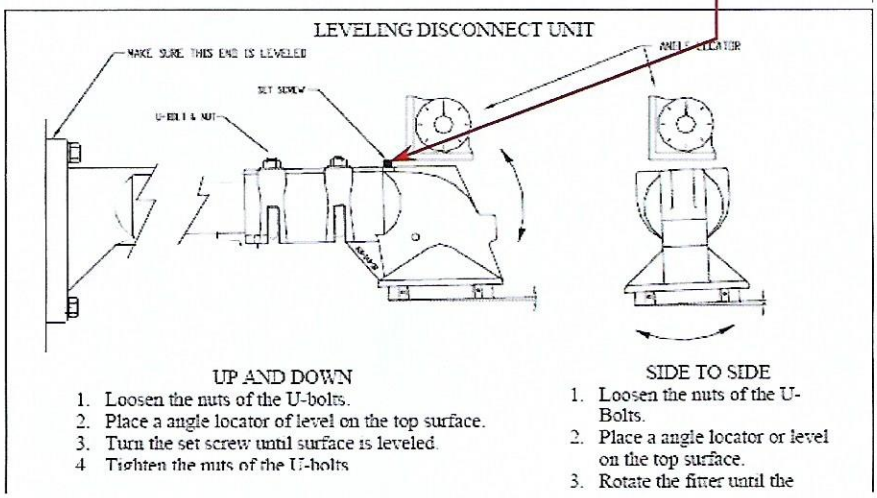
Never operate the system over people. For safety precautions, people should stay clear of the assembly during the raising and lowering operation.

CAMERA LOWERING SYSTEMS

LEVELING POINTS

PROVIDES ABILITY TO LEVEL DISCONNECT UNIT FROM SIDE TO SIDE AND FRONT TO BACK ON NON-PLUMB POLES

Leveling setscrew with locknut to accommodate for front to back adjustment of non-plumb pole.

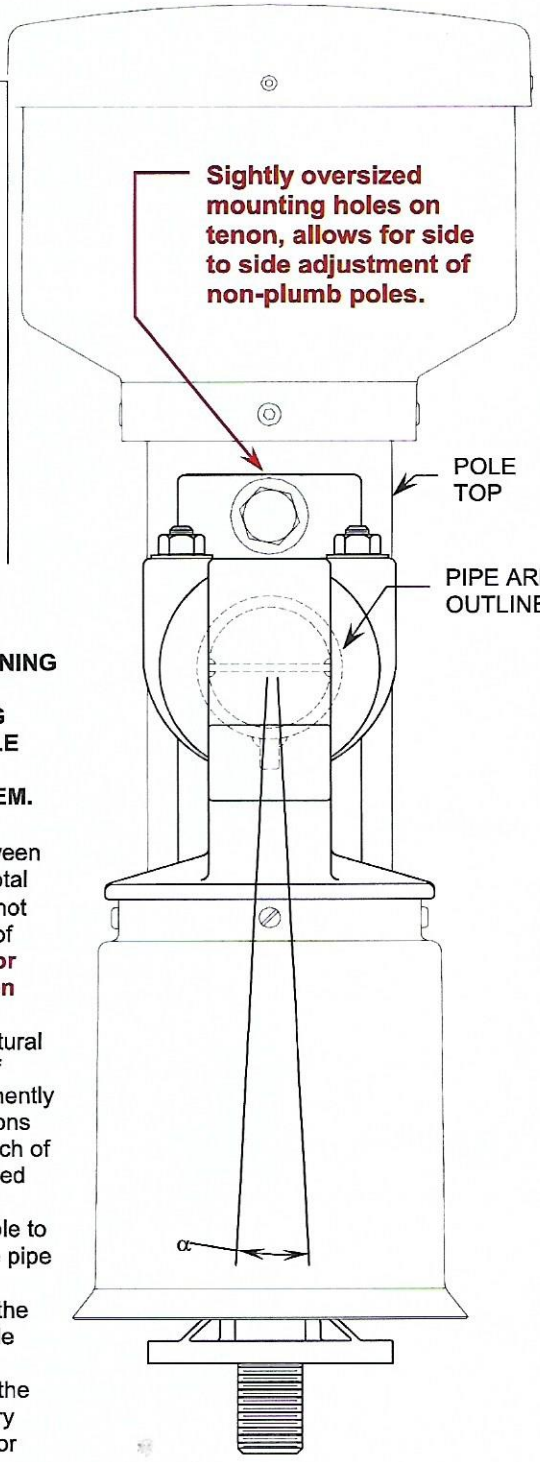


UP AND DOWN

1. Loosen the nuts of the U-bolts.
2. Place a angle locator of level on the top surface.
3. Turn the set screw until surface is leveled.
4. Tighten the nuts of the U-bolts

SIDE TO SIDE

1. Loosen the nuts of the U-Bolts.
2. Place a angle locator or level on the top surface.
3. Rotate the fitter until the



END VIEW

DETAILS OF FEATURES

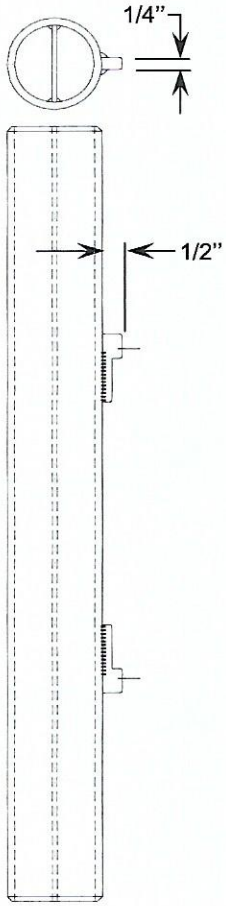


Fig. 1

NOTE: **WHEN THE INTERLOCKING POSITIONING KEYS OF THE ARM ASSEMBLY ARE MATED WITH THE CORRESPONDING NOTCHES IN THE FITTERS, THE POLE SHAFT MUST BE PLUMB FOR THE PROPER OPERATION OF THE SYSTEM.**

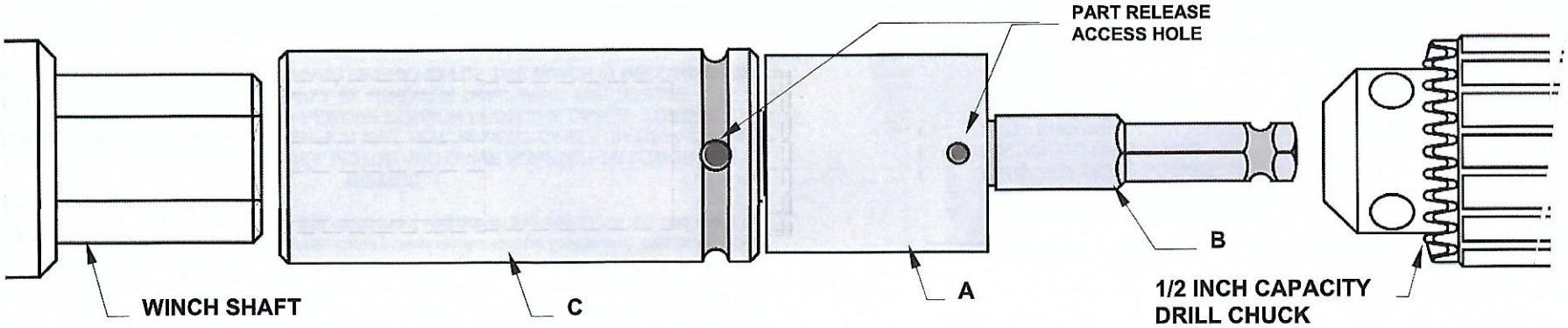
ANGLE α: The angle α shown in the END VIEW is based on mechanical tolerances between mating parts and should not exceed a total of 1/2 °. This deviation from plumb will not affect the operation of the components of the arm assembly. **The angle allows for side to side adjustments needed when pole is not plumb.**

PIPE ARM: **(See Fig. 1)** Constructed of 2 inch structural steel pipe having an outside diameter of 2-3/8 inch. Positioning keys are permanently welded to the pipe arm at precise positions that align with notches in the ends of each of the fitters. Arm finish is hot dip galvanized after all welding is completed. Optional finishes over the galvanizing are available to match the color of the pole. Ends of the pipe arm bottom out against the inside of the fitters a small fraction of an inch before the keys bottom out in the notches to provide a secure fit.
The pipe arm is installed complete with the rest of the arm components at the factory and is pre-wired to eliminate any need for adjustment in the field.

DRILL MOTOR TO WINCH DRIVE ADAPTOR WITH CLUTCH

CAMERA LOWERING SYSTEMS

ASSEMBLY AND USE INSTRUCTIONS CATALOG NO. ACC-100-DRAD



WARNING: TO REDUCE THE RISK OF PERSONAL INJURY AND TO AVOID SYSTEM DAMAGE, DO NOT OPERATE THE DRIVE ADAPTOR AT MORE THAN 350 RPM. ***WEAR SAFETY GLASSES AT ALL TIMES DURING THE OPERATION OF THIS DEVICE.***

NOTES:

- DRIVE ADAPTOR MUST BE USED WITH A DRILL MOTOR HAVING AT LEAST A 1/2 INCH CAPACITY CHUCK, VARIABLE SPEED, REVERSIBLE, AND SHOULD BE RATED FOR 350 RPM (MAX. OPERATING SPEED) OR LESS.
- THE PURPOSE OF THIS DRIVE ADAPTOR IS TO PROVIDE A SAFE AND EFFICIENT MEANS OF RAISING AND LOWERING A POLE MOUNTED LUMINAIRE WITH THE ASSISTANCE OF A HAND HELD DRILL MOTOR. AN AUTOMATICALLY RESETTABLE TORQUE LIMITER (OVERLOAD CLUTCH) IS USED TO PROTECT THE OPERATOR AND THE SYSTEM FROM POTENTIALLY DAMAGING AMOUNTS OF TORQUE.
- THE HANDLE ASSEMBLY FOR THE HAND OPERATION OF THE WINCH SHOULD BE KEPT NEARBY AS IT IS NEEDED FOR THE LOCKING AND UNLOCKING OF THE ELECTRICAL CONTACT UNIT. FOLLOW THE PROCEDURE SPECIFIED WITHIN THESE INSTRUCTIONS.

CAUTION:

THE TORQUE LIMITER OF THIS DRIVE ADAPTOR IS INTENDED FOR THE EXPRESS LIFTING AND LOWERING OF A POLE MOUNTED LUMINAIRE. DO NOT USE THIS DRIVE ADAPTOR FOR ANY OTHER LIFTING OR PULLING PURPOSES NOT SPECIFIED WITHIN THESE INSTRUCTIONS.

THE PARTS OF THIS DRIVE ADAPTOR ARE PRE-ASSEMBLED AT THE FACTORY AND HAS A PRESET TORQUE OF APPROXIMATELY 240 lb.in. SHOULD IT BE NECESSARY TO DISASSEMBLE THE PARTS, USE A SMALL DIAMETER BLUNT TIPPED ROD OR PIN IN THE PART RELEASE ACCESS HOLE OF THE PART TO BE REMOVED, AND PUSH THE SPRING LOADED PIN IN UNTIL THE PARTS CAN BE PULLED APART.

IT SHOULD NOT BE NECESSARY TO ADJUST THE TORQUE SETTING ON THE TORQUE LIMITER (**ITEM A**), HOWEVER, SHOULD ADJUSTMENT BE REQUIRED, FOLLOW THE INSTRUCTIONS ON THE SEPARATE SHEET SPECIFICALLY FOR THE TORQUE LIMITER MODEL 12 B. TORQUE RANGE: MIN. 240 lb. in.(27.1 Nm.) – MAX. 600 lb. in. (67.8 Nm.)

TO UNLOCK AND LOWER LUMINAIRE

1. Using the handle for the manual operation of the winch, operate the winch to raise the luminaire approximately 3/4 in. This will unlock the electrical disconnect unit. Operate the winch by hand to lower the luminaire enough to confirm that the luminaire is separated from the disconnect unit. Remove the manual use handle from the winch.
2. Insert the drill motor adaptor shaft (**ITEM C**) of the drive adaptor assembly into the chuck of the drill and tighten the chuck. Slip the 1-1/8 in. hex drive socket (**ITEM D**) of the drive adaptor over hex shaft of the winch until shaft is fully inserted into the socket. Keeping the drill and the drive adaptor in line with the winch shaft, operate the drill to lower the luminaire to the desired height for servicing. Be sure to keep drive socket over the winch shaft at all times during drill operation.

CAUTION:

SHOULD THE DRIVE SOCKET SLIP OFF OF THE WINCH SHAFT DURING LOWERING OR RAISING OPERATIONS, DO NOT ATTEMPT TO PUT DRIVE SOCKET BACK ON THE WINCH SHAFT UNTIL ALL PARTS HAVE STOPPED MOVING.

TO RAISE AND LOCK LUMINAIRE

3. After servicing the luminaire, slip the hex drive socket of the drive adaptor over the winch shaft completely and operate the drill at a moderate speed to raise the luminaire to a position that will place the moveable portion of the disconnect unit about one foot from engaging the fixed portion, and then stop the raising operation. To minimize pole shaking and vibrations it may be necessary to stop the operation periodically during the raising of the luminaire until movement subsides.

(continued on sheet 2 of 2)

DRILL MOTOR TO WINCH DRIVE ADAPTOR W/CLUTCH

CAMERA LOWERING SYSTEMS

ASSEMBLY AND USE INSTRUCTIONS CATALOG NO. ACC-100-DRAD

TO RAISE AND LOCK LUMINAIRE (Continued)

WARNING:

DO NOT ATTEMPT TO RAISE THE LUMINAIRE INTO THE ELECTRICAL DISCONNECT UNIT WITHOUT STOPPING PRIOR TO ENGAGEMENT. AN EXCESSIVE AMOUNT OF TORQUE IS GENERATED WHEN THE LUMINAIRE AND ATTACHED GUIDE POST SPIN AROUND INTO POSITION WHEN MATING WITH THE FIXED PORTION OF THE DISCONNECT UNIT. TORQUE IS COMPOUNDED BY THE SIZE, WEIGHT, AND SHAPE OF THE LUMINAIRE. ENGAGEMENT OF THE MOVEABLE PORTION OF THE DISCONNECT UNIT (ATTACHED TO THE TOP OF THE LUMINAIRE) WITH THE FIXED PORTION OF THE DISCONNECT UNIT (ATTACHED TO END OF POLE MOUNTED ARM) SHOULD BE DONE SLOWLY TO CONTROL POTENTIALLY DAMAGING TORQUE.

4. Proceed to *SLOWLY* raise the luminaire by operating the drill motor in the raising direction at its slowest speed. The luminaire will noticeably turn as the disconnect unit is aligning for engagement. Continue to slowly raise the luminaire. When all parts are in proper position and at the very top, the raising motion will stop and an audible clicking will be coming from the torque limiter as the limiter slips. Additionally, noticeable jerking motion can be felt by the operator. Immediately stop the drill motor when the sound and jerking motion are sensed. Failure to stop the drill motor immediately will place unnecessary wear on the torque limiter and lead to reduced accuracy. At this stage, the luminaire and disconnect unit are all the way to the top and there is still tension on the cable.

NOTICE:

ATTEMPTING TO REVERSE THE DRILL MOTOR AND DRIVE ADAPTOR TO LOWER THE LUMINAIRE INTO THE 'LOCKED' POSITION WILL NOT, IN MOST CASES, SATISFACTORILY LOCK THE DISCONNECT UNIT AND REMOVE TENSION FROM THE CABLE. TORQUE REQUIRED FOR THIS OPERATION WILL BE GREATER THAN WHAT THE TORQUE LIMITER IS INTENDED FOR. ONLY HAND OPERATION OF THE WINCH IS RECOMMENDED.

5. Remove the drill motor and the drive adaptor from the winch shaft. Install the manually operated winch handle onto the winch shaft and secure. Operate the winch to lower the luminaire approximately 3/4 inch. Continue to operate the winch in the lowering direction until there is noticeable slack in the control cable, indicating that the luminaire and disconnect unit are in the locked position.
6. For additional information on the lowering system and the winch, refer to specific instructions for those items.

CAUTION:

ALWAYS BE SURE THAT THE SPRING ACTIVATED RETAINING PINS OF THE DRIVE COMPONENTS ARE ENGAGED WITH MATING PARTS PRIOR TO OPERATION OF THE DRIVE ADAPTOR.

MAINTENANCE AND CARE OF DRIVE ADAPTOR

- PERIODICALLY EXAMINE ALL MECHANICAL COMPONENTS FOR EXCESSIVE WEAR.
- REPLACE ANY PARTS THAT EXHIBIT EXCESSIVE WEAR OR VISIBLE DAMAGE SUCH AS CRACKS OR BENT SHAFTS.
- PROTECT PARTS FROM DIRECT CONTACT WITH WATER OR ANY OTHER CORROSIVE LIQUIDS.
- TORQUE LIMITER CONTAINS LUBRICATED PARTS. EXCESSIVELY HOT OR COLD TEMPERATURES MAY AFFECT THE ACCURACY OF THE TORQUE SETTING.
- CHECK ALL RETAINING PINS OF THE DRIVE COMPONENTS FOR PROPER OPERATION.

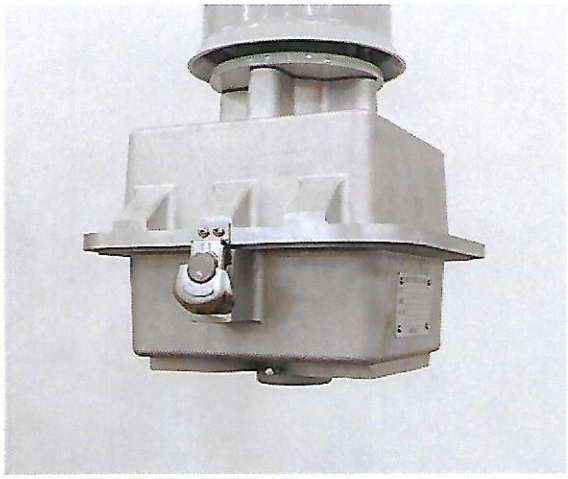
PARTS LIST

	Description	Qty.	Part Number
A.	ADJUSTABLE TORQUE LIMITER	1 EA.	12 B
B.	1/2" TO DRILL MOTOR ADAPTOR	1 EA.	M-404
C.	1-1/8" HEX DRIVE SOCKET	1 EA.	670-102-00
	RETAINING RING PLIERS (NOT SHOWN)	1 EA.	Z-4

For replacement parts, contact:
Camera Lowering Systems at 708-681-4330
www.lowering-device.com

CAMERA LOWERING SYSTEMS

FLA-110 FLANGE INSTALLATION GUIDE

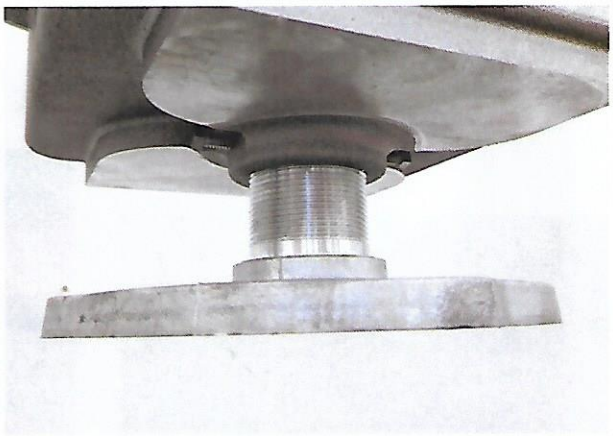


Camera Junction Box (CAMJB-10)

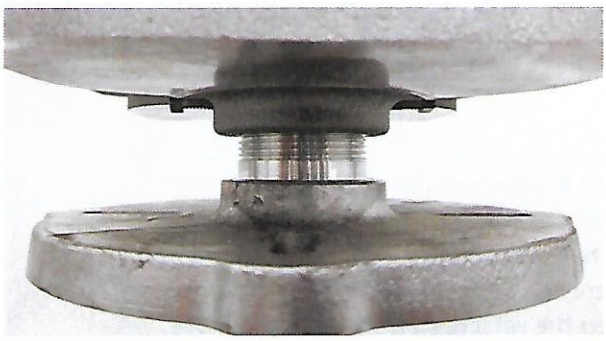
Camera to Camera Junction Box Flange (FLA-110)

STEP 1: Back out the two set screws on the 1.5" conduit thread, on the bottom of the Camera Junction Box.

Set screw backed out so as to not interfere with the threading of the flange.

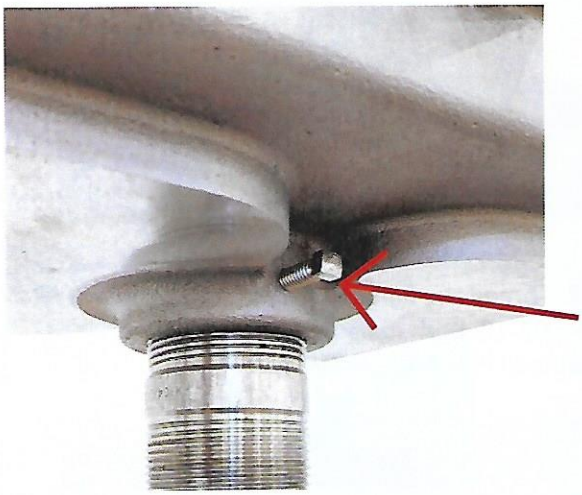


STEP 2: Thread the flange onto the bottom of the CAMJB-10. Both male and female threads are National Pipe Straight (NPS).



STEP 3: Thread until it stops. You should be able to thread 1" of threading.

1" of threading was accomplished onto the CAMJB-10.



STEP 4: After the flange is threaded on, apply Loctite 271 Threadlocker (or equal) on to the setscrews. Tighten the setscrews evenly on both sides. The use of loctite is to keep the setscrews from vibrating loose.

CAMERA LOWERING SYSTEMS

LIGHTING & LOWERING SYSTEMS®

PREVENTATIVE MAINTENANCE

Upon installation of the system, periodic preventive maintenance should be done to the system. The following is a list of preventative maintenance operations and schedule:

LOWERING TOOL

1. **The tool should be kept in a clean and dry area.**
2. **During every usage of the Lowering Tool (LT-CC, LT-4), the raising and lowering cable should be checked for kinks, cut strands, and any irregularities, each time the tool is used. Do not run cable thru bare hands, as**
3. **The gears on the winch should be checked for gearing grease. If the gear box looks dry, gear grease should be applied on the gears only, with a small paint brush. The grease should be applied liberally covering all gears. Recommended grease should be UNIWRI 2 product #C163520 Manufactured by: Fuchs Lubricants. (PH: 800-800-OILS) This grease can be obtained through Camera Lowering Systems.**
4. **A drop of 10W-30 oil should be applied in opening of casement (see sign on winch). Only apply one drop annually. After applying, crank the winch at least 3 revolutions to distribute the oil.**
5. **Check the cable for any kinks, bends, or stray cut wires. This will tell you if cable is rubbing on an obstruction. If cable is damaged, it will weaken the cable, possibly causing the cable to break. Damaged cables should be replaced. See attached "cable handling" page. Handle wire rope with gloves to avoid possible hand cuts caused by stray wires.**

DISCONNECT UNIT

1. **Each time the disconnect unit is lowered, check the condition of the system. If there are any signs of irregularities, such as scrapes, contact the factory.**
2. **Keep the guide pin cleaned and well lubricated with: 'Super Lube multipurpose grease with Syncolon, manufactured by SYNCO CHEMICAL CORP' or equivalent.**
3. **The locking cams should be checked each time the camera is lowered. Ensure shoulder screw is tightened. Add lubrication if needed on locking cam "rest button".**
4. **Lubrication should be "Super Lube" Multi-purpose grease with Syncolon® or equivalent heavy duty synthetic grease with temperature rating of -45ºF to +450ºF.**
5. **Check all screws, nuts and fastenings are tight.**
6. **Check composite cable for any irregularities.**
7. **The Camera Junction Box should be bolted to the disconnect unit using the four (4) ½-20 Hex screws with lockwashers provided by CLS. Torque Ratings: Torque rating on all ¼" bolts should be 20-25 LBXFT.**
8. **CAMJB-10 set screws, LOCTITE RED is needed to secure the screws from vibration.**

GENERAL NOTES AND COMMENTS

Our raising and lowering system, using the **Individual Disconnect Unit (IDU)**, incorporates the locked suspension, which ensures the full weight of the camera is carried by the IDU, relieving the lowering tool, pulleys, and wire rope from tension. These in fact, only come under tension during the actual raising and lowering operations. It should be borne in mind, however, that even when the IDU is in the locked position, the lowering cable is attached to the lowering tool or the eyebolt, giving the unit a secondary suspension.

When the electrical connection is broken during unlocking and lowering, no live electrical wires are lowered with the camera.

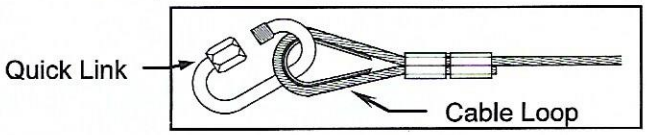
CAUTION:

1. The camera should not be lowered during heavy winds.
2. **DO NOT STAND UNDER CAMERA DURING THE RAISING AND LOWERING OPERATIONS.**

LOWERING THE CAMERA

1. Read all instructions. If you do not understand any part of it, call your Sales Rep. or the factory, at 1 (800) 229-4330.
2. Bring the Portable Lowering Tool and drill to the pole.
3. With poles that have 2 service handholes, remove handhole covers from both handholes. If the pole has only one large handhole, remove the cover and set it aside. Do not misplace handhole cover screws.
4. Insert the Lowering Tool into the bottom handhole for a pole with 2 handholes (or the bottom of the handhole opening for a pole with one large handhole. Make sure that the handhole bracket clip "L" shaped piece catches underneath the inside of the handhole frame. (see the sketch).

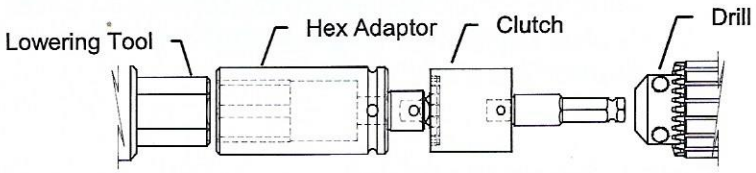
5. The bottom of the handhole has a ½"-13 tapped hole. Bolt the Lowering tool to the bottom of the handhole using the ½" bolt provided. This will stabilize the Lowering Tool to the pole.
6. Open the communications cabinet and check for electrical continuity of the camera.
7. Manually un-reel about 18 to 24" of cable from the winch drum.
8. Reach into upper handhole (for a pole with 2 handholes) and grab the cable that is connected to the camera being lowered. The cable should be attached to an eyebolt inside the pole.
9. Connect the lowering tool cable quick-link to the camera cable loop.
10. Close the quick-link to keep the lowering tool cable from slipping out.



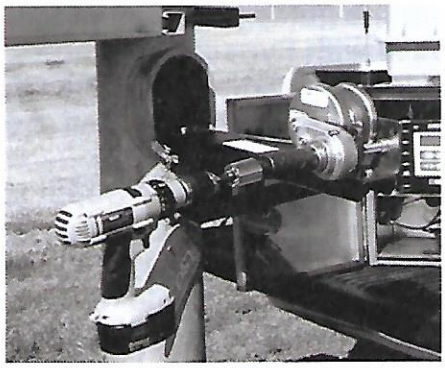
11. Disengage the camera-cable quick-link from the eye bolt in the pole.
12. Manually crank the lowering tool, winding the loose cable on the drum. When the loose cable has been wound up on the drum, continue cranking (or raising) until extra resistance is felt. During the unlocking process, the camera will raise ¾" and then unlocks from the arm.

WARNING: DO NOT CONTINUE TRYING TO RAISE THE CAMERA WHEN ADDITIONAL RESISTANCE IS FELT because the cable will become tighter and tighter. As this occurs, damage to the cable or the Lowering Tool is possible, resulting in the camera disconnecting and falling to the ground.

13. After the camera is raised, reverse the cranking direction and manually lower the camera. If there is slack on the cable during the lowering process, repeat 12 above.
14. If a MANUAL lowering operation is used, continue cranking camera down until camera is about 4ft from the ground or until the operator sees slack on the drum cable. Slack on the drum cable means that the cable quick-link has reached the arm assembly mounted to the pole. The quick-link will not pass through the pulley assembly. The manual lowering operation is complete at this time.
15. If the DRILL ASSEMBLY is used, manually crank the camera and lower it 4ft from the top of the pole. Then the drill assembly can be used.
16. Make sure the drill direction is positioned for lowering the camera.
17. Remove the winch handle.



18. Attach the drill with clutch and adaptor, to the hex head on the winch (see drawing above). Although the Torque Limit is set at the factory, the torque limit may have to be field set.

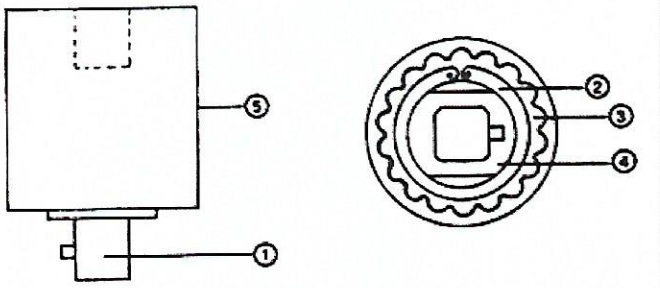


19. Lower the camera down (do not exceed 300 RPM with the drill) until the camera is about 4ft from the ground or until the operator sees slack on the cable. The lowering operation is complete at this time. **NOTE: USE THE DRILL PROVIDED BY *CAMERA LOWERING SYSTEMS* ONLY.** Higher revolutions will limit the life of the brake in

the winch. The brake keeps the camera from free falling during raising and lowering operations.

Torque Limiter

- Attach any calibrated torque indicator to output stub (1) and determine present torque setting while holding the body (5), or vice-versa.
- Remove snap ring (2) and locking plate (3).
- Adjust nut (4) with open-end wrench — clockwise to increase torque, counter-clockwise to decrease torque.
- Obtain new torque reading with the calibrated torque indicator. Repeat preceding step if more adjustment is necessary to reach desired limit.
- Replace locking plate into notches and install snap ring. If locking plate does not "seat," move the adjusting nut slightly until it drops in place. The direction is best determined by whether a minimum torque application or a maximum one is desired.



RAISING THE CAMERA

20. While in the lowered position, check the Disconnect Unit for any irregularities. If there are any signs of irregularities, contact the factory 1 (800) 229-4330. See the Preventative Maintenance section at the end of this document for other items to check.
21. Before raising the camera, a "jumper cable" can be used for connecting the Disconnect Unit to the cabinet, in order to test the camera.
22. After completing the maintenance of the camera, raise the camera to within 4ft of the top using the drill or the crank handle. **Do not exceed 300 RPM with the drill.**
23. If using the drill, remove the drill assembly and put the manual crank handle on to raise the final 4ft.
24. Crank at a medium to slow pace until the camera reaches the top. The camera may rotate up to 180° clockwise or counter clockwise.
25. When the camera reaches the top and the operator feels like the camera cannot be cranked up any further, **STOP** cranking upwards and slowly reverse the cranking direction. **DO NOT OVERCRANK!**
26. The reversed cranking direction will lower the camera into the locked or resting position. The operator should see slack on the cable and on the lowering tool after about ½ of a crank revolution. Check the communications cabinet. The camera should be energized.
27. With the slack on the cable and the camera cable still attached to the Lowering Tool cable, attach the camera cable quick link, to the eyebolt in the pole.
28. After the cable is attached to the eyebolt and the connecting quick-link is closed, disengage the Lowering Tool cable from the camera's cable. **Two quick links should always be used during the unlocking and locking procedures.**
29. Loosen the outside handhole bracket on the lowering tool and push the Lowering Tool slightly into the pole.
30. Lift up and remove the Lowering Tool from the lower handhole.
31. Install the handhole covers back on the pole.
32. Close and secure the communications cabinet.

Preventive Maintenance

Upon installation of the lowering system, periodic preventive maintenance should be performed. The following is a list of preventive maintenance operations and schedule:

LOWERING TOOL

1. **The tool should be kept in a clean and dry area.**
2. **During <u>every</u> use of the Lowering Tool, the raising and lowering cable should be checked for kinks, cut strands, and any irregularities. Do not run the cable using bare hands, as strayed strands could cause cuts and injuries to hands.**
3. **The gears on the winch should be checked for gearing grease. If the gear box looks dry, gear grease should be applied on the gears only, with a small paint brush. The grease should be applied liberally to cover all gears. Recommended grease should be UNIWRI 2 product #C163520 Manufactured by: Fuchs Lubricants. (Phone: 800-800-OILS) This grease may be obtained through *Camera Lowering Systems*.**
4. **A drop of 10W-30 oil should be applied in opening of casement (see label on the winch). Only apply one drop annually. After applying, crank the winch at least 3 revolutions to distribute the oil.**
5. **Check the cable for any kinks, bends, or strayed cut wires. This will tell you if the cable is rubbing on an obstruction. If the cable is damaged, it will be weakened and possibly break. Damaged cables should be replaced. Handle the wire rope with gloves to avoid possible hand cuts caused by strayed wires.**

DISCONNECT UNIT

1. **Each time the Disconnect Unit is lowered, check the condition of the system. If there are any signs of irregularities (such as bent parts, missing hardware, or frayed cable) contact the factory.**
2. **Keep the guide pin greased and cleaned. Brush on *Super Lube*® by *Synco Chemical* (can be purchased through *Camera Lowering Systems*).**
3. **The locking cams should be checked each time the camera is lowered. Ensure that the shoulder screw is tightened. Check all screws, nuts, and fastenings to ensure that they are tight.**
4. **Check the composite cable for any irregularities.**

If the system does not operate properly, repeat steps 1-32 two more times, adjusting the torque on the torque limiter if necessary. The torque model is the 12A. It is set at the factory at the Min. setting of 240 lb./in. (27.1 Nm.). The max setting is 600 lb./in. (67.8 Nm.).

Total Length:	319 ft
Number of Reports:	3
Number of Passing Reports:	3
Number of Failing Reports:	0
Number of Warning Reports:	0
Documentation Only:	0

LINKWARE™ PC
CABLE TEST MANAGEMENT SOFTWARE

Cable ID: 261220-6A-15M28M7042-01

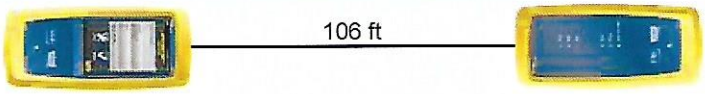
Test Summary: PASS

Test Limit: ISO11801 PL Class D
Limits Version: V7.11
Date / Time: 12/05/2025 02:42:46 PM
Operator: MARTIN FLORES
Headroom 0.9 dB (NEXT 4,5-7,8)
Cable Type: Cat 6A U/UTP
NVP: 70.0%

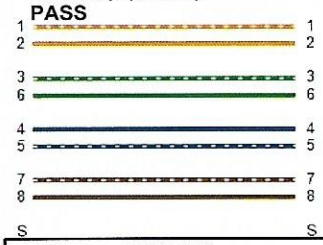
Main: Versiv
S/N: 18230413
Software Version: V6.13 Build 2
Calibration Date: 12/26/2024
Adapter: DSX-5000 (DSX-CHA004)
S/N: 4232051

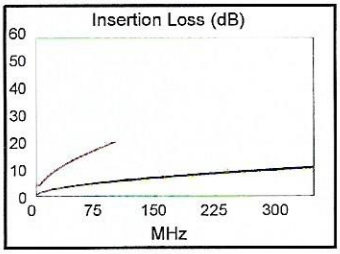
Remote: Versiv
S/N: 18230412
Software Version: V6.13 Build 2
Calibration Date: 12/26/2024
Adapter: DSX-5000R (DSX-CHA004)
S/N: 4232031

Length (ft), Limit 295	[Pair 1,2]	106
Prop. Delay (ns), Limit 498	[Pair 3,6]	161
Delay Skew (ns), Limit 44	[Pair 3,6]	7
Resistance (ohms), Limit 21.00	[Pair 3,6]	4.70
Insertion Loss Margin (dB)	[Pair 3,6]	14.8
Frequency (MHz)	[Pair 3,6]	100.0
Limit (dB)	[Pair 3,6]	20.4

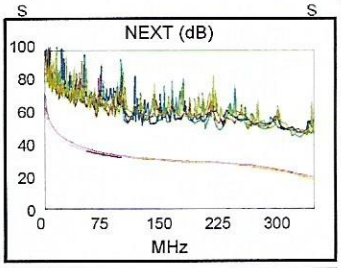


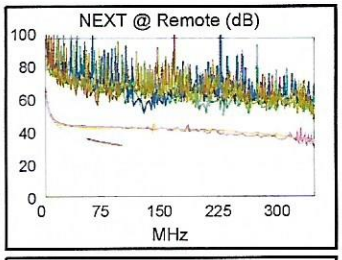
Wire Map (T568B)
PASS



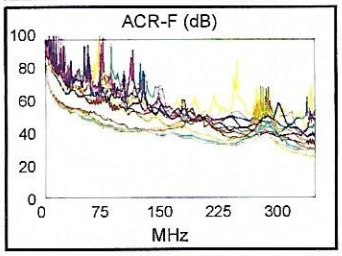


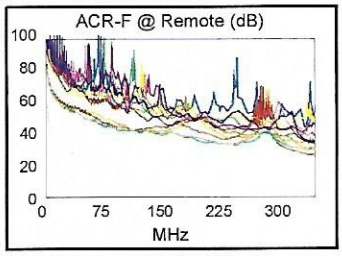
	Worst Case Margin		Worst Case Value	
PASS	MAIN	SR	MAIN	SR
Worst Pair	4,5-7,8	1,2-3,6	1,2-3,6	4,5-7,8
NEXT (dB)	0.9	7.0	1.2	10.2
Freq. (MHz)	95.3	52.0	100.0	94.3
Limit (dB)	32.7	37.0	32.3	32.7
Worst Pair	4,5	3,6	1,2	4,5
PS NEXT (dB)	3.9	10.0	4.2	13.2
Freq. (MHz)	95.3	52.0	100.0	94.3
Limit (dB)	29.7	34.0	29.3	29.7



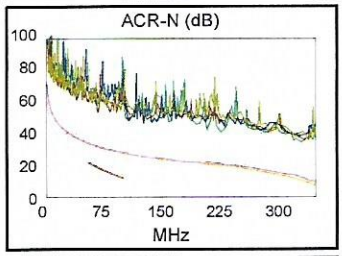


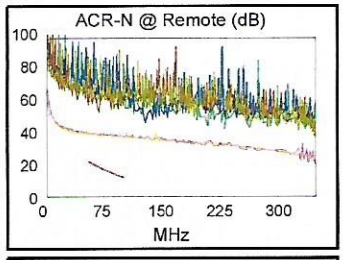
PASS	MAIN	SR	MAIN	SR
Worst Pair	7,8-4,5	4,5-7,8	7,8-4,5	4,5-7,8
ACR-F (dB)	22.7	22.9	22.8	23.0
Freq. (MHz)	97.3	97.3	98.0	98.0
Limit (dB)	18.9	18.9	18.8	18.8
Worst Pair	4,5	7,8	4,5	7,8
PS ACR-F (dB)	25.6	25.7	25.7	25.7
Freq. (MHz)	97.3	97.3	98.0	97.3
Limit (dB)	15.9	15.9	15.8	15.9



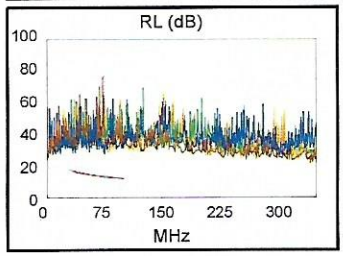


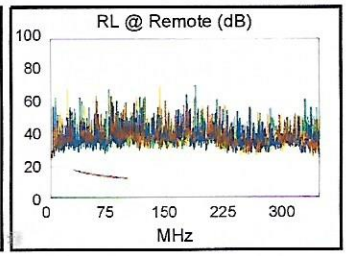
PASS	MAIN	SR	MAIN	SR
Worst Pair	1,2-3,6	1,2-3,6	1,2-3,6	1,2-3,6
ACR-N (dB)	11.7	17.0	16.0	25.7
Freq. (MHz)	50.0	50.0	100.0	99.5
Limit (dB)	23.2	23.2	11.9	12.0
Worst Pair	3,6	3,6	3,6	4,5
PS ACR-N (dB)	14.7	20.0	19.0	27.5
Freq. (MHz)	50.0	50.0	100.0	94.3
Limit (dB)	20.2	20.2	8.9	10.0





PASS	MAIN	SR	MAIN	SR
Worst Pair	4,5	3,6	4,5	3,6
RL (dB)	12.9	13.4	15.1	13.4
Freq. (MHz)	33.0	35.8	74.5	35.8
Limit (dB)	16.8	16.5	13.3	16.5





Compliant Network Standards:

10BASE-T	100BASE-TX	100BASE-T4
1000BASE-T	2.5GBASE-T	ATM-25
ATM-51	ATM-155	100VG-AnyLan
TR-4	TR-16 Active	TR-16 Passive

LinkWare™ PC Version 11.3

Project: DEFAULT
Site: Not Set
Floor: Not Set
Rack: Not Set
Untitled1

Building: Not Set
Room: Not Set
Patch: Not Set

FLUKE networks

LINKWARE™ PC
CABLE TEST MANAGEMENT SOFTWARE

Cable ID: 261220-6A-15M28M7042-02

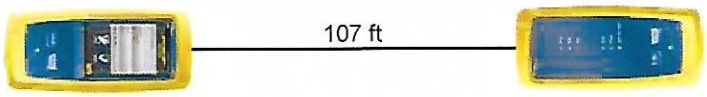
Test Summary: PASS

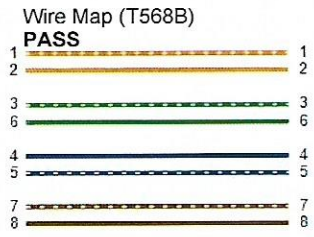
Test Limit: ISO11801 PL Class D
Limits Version: V7.11
Date / Time: 12/05/2025 02:43:31 PM
Operator: MARTIN FLORES
Headroom 0.5 dB (NEXT 4,5-7,8)
Cable Type: Cat 6A U/UTP
NVP: 70.0%

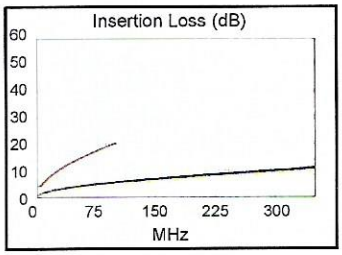
Main: Versiv
S/N: 18230413
Software Version: V6.13 Build 2
Calibration Date: 12/26/2024
Adapter: DSX-5000 (DSX-CHA004)
S/N: 4232051

Remote: Versiv
S/N: 18230412
Software Version: V6.13 Build 2
Calibration Date: 12/26/2024
Adapter: DSX-5000R (DSX-CHA004)
S/N: 4232031

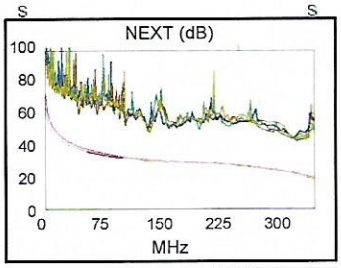
Length (ft), Limit 295	[Pair 1,2]	107
Prop. Delay (ns), Limit 498	[Pair 3,6]	163
Delay Skew (ns), Limit 44	[Pair 3,6]	8
Resistance (ohms), Limit 21.00	[Pair 3,6]	4.70
Insertion Loss Margin (dB)	[Pair 3,6]	14.7
Frequency (MHz)	[Pair 3,6]	100.0
Limit (dB)	[Pair 3,6]	20.4

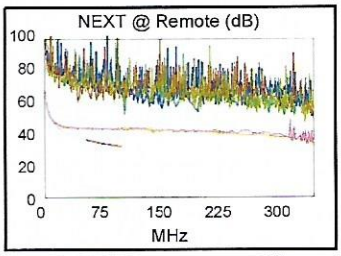




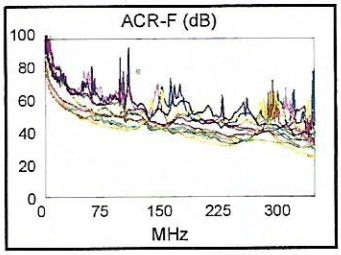


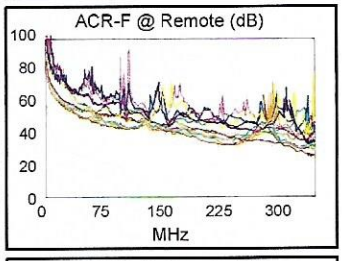
	Worst Case Margin		Worst Case Value	
PASS	MAIN	SR	MAIN	SR
Worst Pair	4,5-7,8	1,2-3,6	4,5-7,8	1,2-3,6
NEXT (dB)	0.5	6.8	0.5	10.4
Freq. (MHz)	97.5	49.8	97.5	97.8
Limit (dB)	32.5	37.3	32.5	32.5
Worst Pair	4,5	3,6	4,5	1,2
PS NEXT (dB)	3.5	9.8	3.5	13.4
Freq. (MHz)	97.5	49.8	97.5	97.8
Limit (dB)	29.5	34.3	29.5	29.5



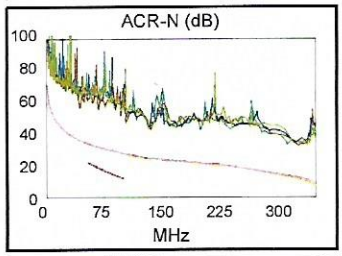


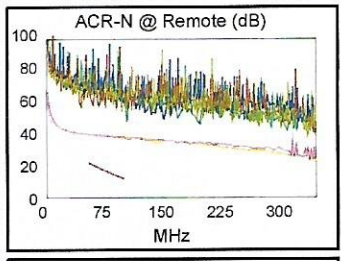
PASS	MAIN	SR	MAIN	SR
Worst Pair	1,2-3,6	3,6-1,2	1,2-3,6	3,6-1,2
ACR-F (dB)	23.5	23.8	24.0	24.4
Freq. (MHz)	82.3	82.3	98.5	98.5
Limit (dB)	20.3	20.3	18.7	18.7
Worst Pair	3,6	1,2	3,6	1,2
PS ACR-F (dB)	25.6	26.3	26.1	26.8
Freq. (MHz)	81.8	91.8	98.3	97.8
Limit (dB)	17.4	16.4	15.8	15.8



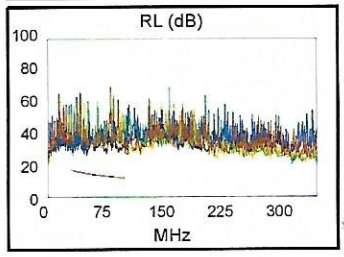


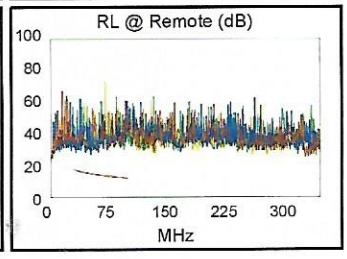
PASS	MAIN	SR	MAIN	SR
Worst Pair	1,2-3,6	1,2-3,6	1,2-3,6	1,2-3,6
ACR-N (dB)	11.3	16.7	15.7	25.0
Freq. (MHz)	49.8	49.0	100.0	97.8
Limit (dB)	23.3	23.5	11.9	12.3
Worst Pair	3,6	3,6	4,5	3,6
PS ACR-N (dB)	14.3	19.7	18.6	28.0
Freq. (MHz)	49.8	49.0	100.0	97.8
Limit (dB)	20.3	20.5	8.9	9.3





PASS	MAIN	SR	MAIN	SR
Worst Pair	4,5	4,5	4,5	4,5
RL (dB)	11.1	12.7	12.2	13.6
Freq. (MHz)	32.5	32.5	52.0	52.0
Limit (dB)	16.9	16.9	14.8	14.8





Compliant Network Standards:

10BASE-T	100BASE-TX	100BASE-T4
1000BASE-T	2.5GBASE-T	ATM-25
ATM-51	ATM-155	100VG-AnyLan
TR-4	TR-16 Active	TR-16 Passive

LinkWare™ PC Version 11.3

Project: DEFAULT
Site: Not Set
Floor: Not Set
Rack: Not Set
Untitled1

Building: Not Set
Room: Not Set
Patch: Not Set

FLUKE networks

LINKWARE™ PC
CABLE TEST MANAGEMENT SOFTWARE

Cable ID: 261220-6A-15M28M7042-03

Test Summary: PASS

Test Limit: ISO11801 PL Class D
Limits Version: V7.11
Date / Time: 12/05/2025 02:44:02 PM
Operator: MARTIN FLORES
Headroom 0.8 dB (NEXT 4,5-7,8)
Cable Type: Cat 6A U/UTP
NVP: 70.0%

Main: Versiv
S/N: 18230413
Software Version: V6.13 Build 2
Calibration Date: 12/26/2024
Adapter: DSX-5000 (DSX-CHA004)
S/N: 4232051

Remote: Versiv
S/N: 18230412
Software Version: V6.13 Build 2
Calibration Date: 12/26/2024
Adapter: DSX-5000R (DSX-CHA004)
S/N: 4232031

Length (ft), Limit 295	[Pair 1,2]	106
Prop. Delay (ns), Limit 498	[Pair 3,6]	161
Delay Skew (ns), Limit 44	[Pair 3,6]	7
Resistance (ohms), Limit 21.00	[Pair 3,6]	4.70
Insertion Loss Margin (dB)	[Pair 3,6]	14.8
Frequency (MHz)	[Pair 3,6]	100.0
Limit (dB)	[Pair 3,6]	20.4

	Worst Case Margin		Worst Case Value	
PASS	MAIN	SR	MAIN	SR
Worst Pair	4,5-7,8	4,5-7,8	4,5-7,8	4,5-7,8
NEXT (dB)	0.8	7.2	1.0	10.3
Freq. (MHz)	97.8	55.3	100.0	97.8
Limit (dB)	32.5	36.5	32.3	32.5
Worst Pair	4,5	4,5	4,5	4,5
PS NEXT (dB)	3.8	9.9	4.0	13.3
Freq. (MHz)	97.8	51.0	100.0	97.8
Limit (dB)	29.5	34.1	29.3	29.5

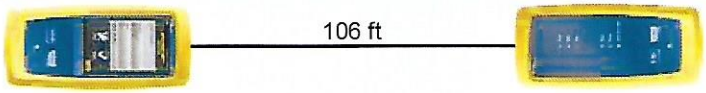
PASS	MAIN	SR	MAIN	SR
Worst Pair	3,6-1,2	1,2-3,6	1,2-3,6	7,8-4,5
ACR-F (dB)	22.8	22.7	24.2	24.0
Freq. (MHz)	4.5	4.5	99.0	98.5
Limit (dB)	45.5	45.5	18.7	18.7
Worst Pair	1,2	1,2	7,8	7,8
PS ACR-F (dB)	24.5	24.9	26.2	26.3
Freq. (MHz)	4.5	6.9	98.5	98.3
Limit (dB)	42.5	38.9	15.7	15.8

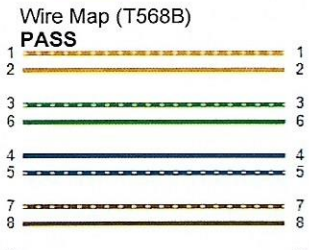
PASS	MAIN	SR	MAIN	SR
Worst Pair	1,2-3,6	1,2-3,6	1,2-3,6	4,5-7,8
ACR-N (dB)	11.4	17.4	15.8	25.2
Freq. (MHz)	49.8	49.8	99.3	97.8
Limit (dB)	23.3	23.3	12.0	12.3
Worst Pair	3,6	4,5	4,5	4,5
PS ACR-N (dB)	14.4	20.1	18.8	28.0
Freq. (MHz)	49.8	51.0	100.0	97.8
Limit (dB)	20.3	19.9	8.9	9.3

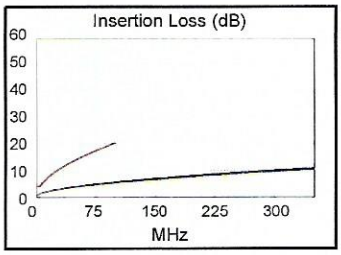
PASS	MAIN	SR	MAIN	SR
Worst Pair	3,6	3,6	1,2	1,2
RL (dB)	12.4	13.1	13.3	16.8
Freq. (MHz)	32.8	32.5	83.8	100.0
Limit (dB)	16.8	16.9	12.8	12.0

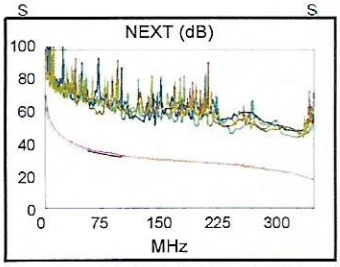
Compliant Network Standards:

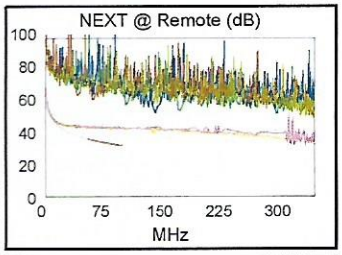
10BASE-T	100BASE-TX	100BASE-T4
1000BASE-T	2.5GBASE-T	ATM-25
ATM-51	ATM-155	100VG-AnyLan
TR-4	TR-16 Active	TR-16 Passive

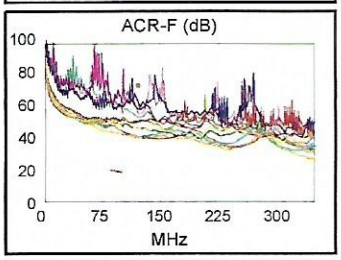


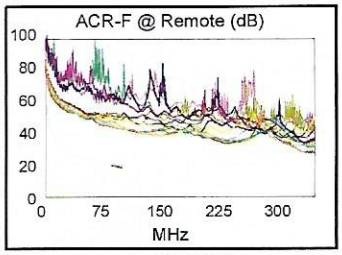


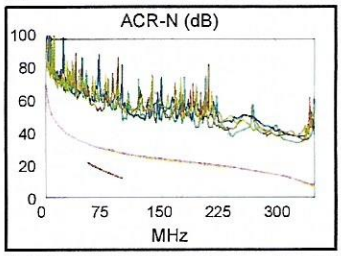


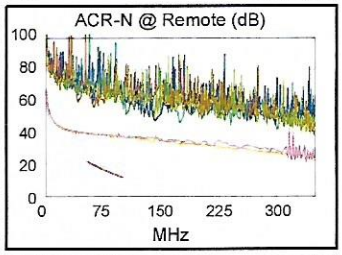


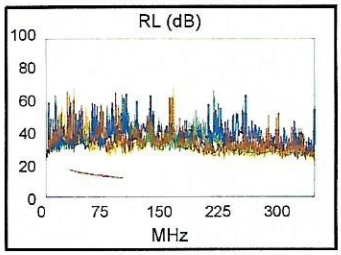


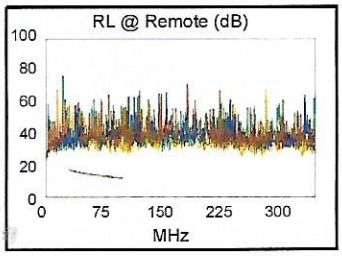












LinkWare™ PC Version 11.3

Project: DEFAULT
Site: Not Set
Floor: Not Set
Rack: Not Set
Untitled1

Building: Not Set
Room: Not Set
Patch: Not Set

FLUKE networks

Thern
Winches & Cranes

EC DECLARATION OF CONFORMITY

Verklaring van overeenstemming / Konformitätserklärung / Declaración de conformidad / Déclaration de conformité

This certificate meets the requirements of the directives listed below.

Dit certificaat voldoet aan de eisen van de onderstaande richtlijnen. / Dieses Zertifikat erfüllt die Anforderungen der unten aufgeführten Richtlinien.

Este certificado cumple con los requisitos de las directivas enumeradas a continuación. / Ce certificat répond aux exigences des directives énumérées ci-dessous.

MANUFACTURER NAME & ADDRESS:
Naam en adres van de fabrikant: / Name und Adresse des Herstellers: / Nombre y dirección del fabricante: / Nom et adresse du fabricant:

Thern, Inc.
5712 Industrial Park Road
Winona, MN 55987
USA
www.thern.com

ADMINISTRATOR OF TECHNICAL FILE:
Beheerder van technisch dossier: / Administrator der technischen Datei: / Administrador de expediente técnico: / Administrateur de dossier technique:

Thern Europe B.V.
Regional Headquarters
Amsterdam Shiphol Tetra
Siriusdreef 17-27, 2132 WT Hoofddorp
The Netherlands

DESCRIPTION OF EQUIPMENT: Beschrijving van apparatuur: / Beschreibung der Ausrüstung: / Descripción del Equipo: / Description de l'équipement:

Hand powered winch for lifting equipment, used with rope assembly meeting requirements
Model M4022PB, M4032PB, M4042PBSS series and cataloged variants and accessories
Rated for lifting up to 453 kg

SERIAL NUMBERS: Serienummers: / Seriennummer: / Número de Serie: / Numéros de série:

Not serialized

DIRECTIVES USED: Gebruikte richtlijnen: / Verwendete Richtlinien: / Directivas utilizadas: / Directives utilisées:

Machinery Directive 2006/42/EC

HARMONIZED STANDARDS USED:
Geharmoniseerde normen gebruikt: / Harmonisierte Standards verwendet: / Estándares armonizados utilizados: / Normes harmonisées utilisées:

EN ISO 12100:2010 – Safety of Machinery – general principles for design; risk assessment

NATIONAL STANDARDS AND TECHNICAL SPECIFICATIONS USED: Gebruikte nationale normen en technische specificaties: / Nationale Normen und technische Spezifikationen verwendet: / Estándares nacionales y especificaciones técnicas utilizadas: / Normes nationales et spécifications techniques utilisées:

None

I declare that the equipment described above meets the essential safety requirements of the European directives listed above.

Ik verklaar dat de hierboven beschreven apparatuur voldoet aan de essentiële veiligheidseisen van de hierboven genoemde Europese richtlijnen.

Ich erkläre, dass die oben beschriebenen Geräte die grundlegenden Sicherheitsanforderungen der oben aufgeführten europäischen Richtlinien erfüllen.

Declaro que el equipo descrito anteriormente cumple con los requisitos de seguridad esenciales de las directivas europeas enumeradas anteriormente.

Je déclare que l'équipement décrit ci-dessus répond aux exigences essentielles de sécurité des directives européennes énumérées ci-dessus.

PLACE OF DECLARATION:
Plaats van aangifte / Ort der Erklärung: / Lugar de declaración: / Lieu de déclaration: Winona, MN USA

DATE OF DECLARATION: (Month DD, YYYY)
Datum aangifte: / Datum der Erklärung: / Fecha de declaración: / Date de déclaration: January 3, 2023

NAME: Jeremy Anderson
Naam: / Name: / Nombre: / Nom:

TITLE: Mechanical Engineer
Titel: / Titel: / Título: / Titre:

SIGNATURE: Handtekening: / Unterschrift: / Firma: / Signature:

Jeremy Anderson

Thern
Winches & Cranes

UK
CA

DECLARATION OF CONFORMITY

In accordance with UK Government Guidance

MANUFACTURER NAME & ADDRESS:	ADMINISTRATOR OF TECHNICAL FILE:
Thern, Inc. 5712 Industrial Park Road Winona, MN 55987 USA www.thern.com	Thern Europe B.V. Regional Headquarters Amsterdam Shiphol Tetra Siriusdreef 17-27, 2132 WT Hoofddorp The Netherlands

DESCRIPTION OF EQUIPMENT:
Hand powered winch for lifting equipment, used with rope assembly meeting requirements
Model M4022PB, M4032PB, M4042PBSS series and cataloged variants and accessories
Rated for lifting up to 453 kg

SERIAL NUMBERS:
Not serialized

DIRECTIVES:
2008 No 1597 Supply of Machinery (Safety) Regulations 2008

HARMONIZED STANDARDS:
EN ISO 12100:2010 – Safety of Machinery – general principles for design; risk assessment

DATE OF DECLARATION: (Month DD, YYYY) January 3, 2023

We declare under our sole responsibility that the equipment described above complies with the essential requirements of the standards listed above.

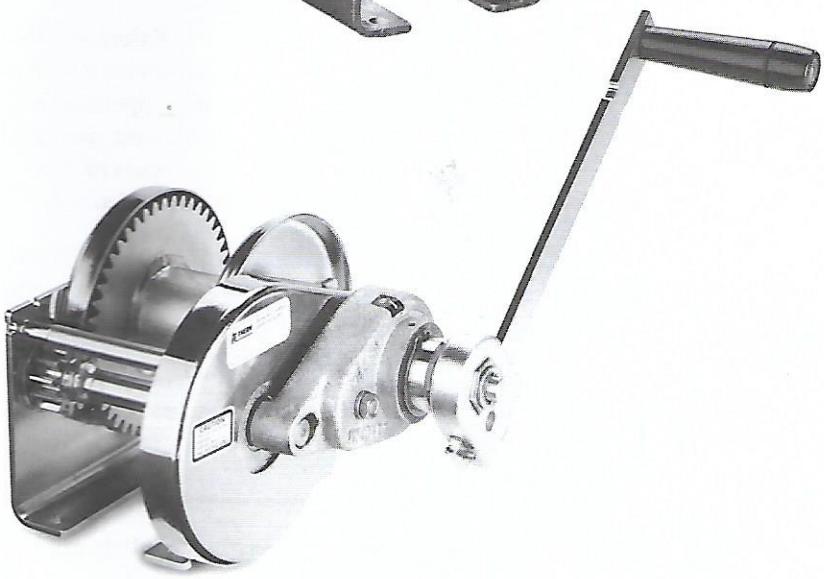
NAME: Jeremy Anderson	**SIGNATURE:** Jeremy Anderson
TITLE: Mechanical Engineer	

Thern
Winches & Cranes

Read this Owner's Manual thoroughly before operating the equipment. Keep it with the equipment at all times. Replacements are available from Thern, Inc., PO Box 347, Winona, MN 55987, 507-454-2996. www.thern.com

IMPORTANT: Please record product information on page 2. This information is required when calling the factory for service.

ORIGINAL TEXT



Owner's Manual

For Thern
Spur Gear Hand Winches

Please record the following:

Date Purchased:

Model No.:

Code No.:

This information is required when calling the factory for service.

Two-Year Limited Warranty

Thern, Inc. warrants its products against defects in material or workmanship for two years from the date of purchase by the original using buyer, or if this date cannot be established, the date the product was sold by Thern, Inc. to the dealer. To make a claim under this warranty, contact the factory for an RGA number. The product must be returned, prepaid, directly to Thern, Inc., 5712 Industrial Park Road, Winona, Minnesota 55987. The following information must accompany the product: the RGA number, the date of purchase, the description of the claimed defect, and a complete explanation of the circumstances involved. If the product is found to be defective, it will be repaired or replaced free of charge, and Thern, Inc. will reimburse the shipping cost within the contiguous USA.

This warranty does not cover any damage due to accident, misuse, abuse, or negligence. Any alteration, repair or modification of the product outside the Thern, Inc. factory shall void this warranty. This warranty does not cover any costs for removal of our product, downtime, or any other incidental or consequential costs or damages resulting from the claimed defects. This warranty does not cover brake discs, wire rope or other wear components, as their life is subject to use conditions which vary between applications.

FACTORY AUTHORIZED REPAIR OR REPLACEMENT AS PROVIDED UNDER THIS WARRANTY IS THE EXCLUSIVE REMEDY TO THE CONSUMER. THERN, INC. SHALL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES FOR BREACH OF ANY EXPRESS OR IMPLIED WARRANTY ON THIS PRODUCT. EXCEPT TO THE EXTENT PROHIBITED BY APPLICABLE LAW, ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE ON THIS PRODUCT IS LIMITED IN DURATION TO THE DURATION OF THIS WARRANTY.

Some states do not allow the exclusion or limitation of incidental or consequential damages, or allow limitations on how long an implied warranty lasts, so the above limitation or exclusion may not apply to you. This warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

Note: Thern, Inc. reserves the right to change the design or discontinue the production of any product without prior notice.

About This Manual

The Occupational Safety and Health Act of 1970 states that it is the employer's responsibility to provide a workplace free of hazard. To this end, all equipment should be installed, operated, and maintained in compliance with applicable trade, industrial, federal, state, and local regulations. It is the equipment owner's responsibility to obtain copies of these regulations and to determine the suitability of the equipment to its intended use.

This Owner's Manual, and warning labels attached to the equipment, are to serve as guidelines for hazard-free installation, operation, and maintenance. They should not be understood to prepare you for every possible situation.

The information contained in this manual is applicable only to the Thern Spur Gear Hand Winches. Do not use this manual as a source of information for any other equipment.

The following symbols are used for emphasis throughout this manual:

⚠WARNING

Failure to follow 'WARNING!' instructions may result in equipment damage, property damage, and/or serious personal injury.

⚠CAUTION

Failure to follow 'CAUTION!' instructions may result in equipment damage, property damage, and/or minor personal injury.

Important!

Failure to follow 'important!' instructions may result in poor performance of the equipment.

Suggestions for Safe Operation

⚠WARNING

DO the following:

Read and comply with the guidelines set forth in this Owner's Manual. Keep this manual, and all labels attached to the winch, readable and with the equipment at all times. Contact Thern, Inc. for replacements.

Check lubrication before use.

Install the wire rope securely to the winch drum.

Keep at least 4 wraps of wire rope wound on the drum at all times, to serve as anchor wraps. With less than 4 wraps on the drum the wire rope could come loose, causing the load to escape.

Keep hands away from the drum, gears, wire rope, and other moving parts of the equipment.

Keep all unnecessary personnel away from winch while in operation. Keep out of the path of the load, and out of the path of a broken wire rope that might snap back and cause injury.

DO NOT do the following:

Do not lift people, or things over people. Do not walk or work under a load or in the line of force of any load.

Do not exceed the load rating of the winch or any other component in the system. To do so could result in failure of the equipment.

Do not use more than one winch to move a load that unless each winch was designed for use in a multiple winch system.

Do not operate with other than manual power.

Do not use damaged or malfunctioning equipment. To do so could result in failure of the equipment.

Do not modify the equipment in any way. To do so could cause equipment failure.

Do not wrap the wire rope around the load. This damages the wire rope and could cause the load to escape. Use approved rigging connectors to secure the wire rope to the load.

Do not operate the winch with guards removed or improperly installed.

Do not divert your attention from the operation. Stay alert to the possibility of accidents, and try to prevent them from happening.

Do not jerk or swing the load. Avoid shock loads by starting and stopping the load smoothly. Shock loads overload the equipment and may cause damage.

Do not lift loads or pull loads on an incline unless the winch is equipped with a brake.

Do not leave a suspended load unattended unless specific precautions have been taken to secure the load and keep people away from the winch and out from under the load.

1.1 Installing the Winch

Important!

- Inspect the winch immediately following installation according to the Instructions for Periodic Inspection. This will give you a record of the condition of the winch with which to compare future inspections.
- A qualified professional should inspect or design the foundation to insure that it will provide adequate support.
- Locate the winch so it will be visible during the entire operation.
- Do not weld the winch frame to the foundation or support structure. Welding the frame may void warranty, contact Thern, Inc. Use fasteners as instructed.

⚠WARNING

Do not install the winch in an area defined as hazardous by the National Electric Code, unless installation in such an area has been thoroughly approved.

Do not install the winch near corrosive chemicals, flammable materials, explosives, or other elements that may damage the winch or injure the operator. Adequately protect the winch and the operator from such elements.

Position the winch so the operator can stand clear of the load, and out of the path of a broken wire rope that could snap back and cause injury.

Attach the winch to a rigid and level foundation that will support the winch and its load under all load conditions, including shock loading.

Install the wire rope securely to the winch drum. A poorly secured wire rope could come loose from its anchor and could release the load. See Figure 3.

1.1.1 CONSULT APPLICABLE CODES AND REGULATIONS for specific rules on installing the equipment.

1.1.2 LOCATE THE WINCH in an area clear of traffic and obstacles that could interfere with operation. Make sure winch is accessible for maintenance and operation.

1.1.3 MAINTAIN A FLEET ANGLE between 1/2 and 1-1/2 degrees. The proper fleet angle minimizes wire rope damage by helping the wire rope wind uniformly onto the drum. See Figure 1.

1.1.4 FASTEN THE WINCH securely to the foundation.

a FOR STANDARD PRODUCTS referred to in this manual, use 3/8 inch coarse thread fasteners, grade 5 or better. Torque for grade 5 fasteners without lubrication is 30 ft. lb. Make sure mounting holes are secured to a solid foundation able to support the winch and the load under all conditions with design factors based on accepted engineering practices.

b NON-STANDARD PRODUCTS that vary from the original design may have different fastening requirements. Contact a structural engineer or Thern, Inc. for this information.

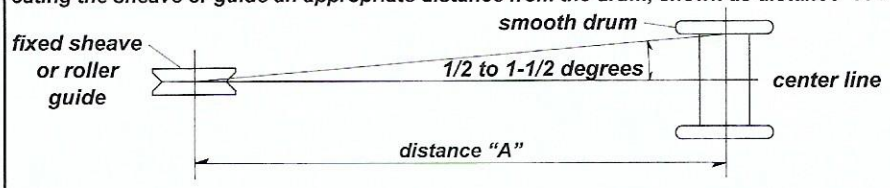
TO COMPLY WITH LOCAL CODES, CONTACT A QUALIFIED PROFESSIONAL TO OBTAIN PROPER STRUCTURE OR FOUNDATION SPECIFICATIONS FOR THE MOUNTING OF THERN PRODUCTS.

Important!

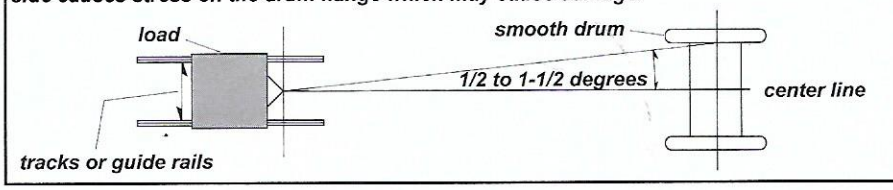
- Use a sheave or roller guide to direct the wire rope to the drum whenever possible.
- Install sheaves, tracks and other equipment so they will remain fixed under all load conditions. Follow the recommendations of the equipment manufacturer.
- Use sheaves of proper diameter to minimize wear on the wire rope. Follow the recommendations of the sheave manufacturer.

Figure 1 – Maintaining the Fleet Angle

When wire rope travels over a sheave or through a roller guide – maintain fleet angle by locating the sheave or guide an appropriate distance from the drum, shown as distance "A".



When wire rope travels directly to the load – maintain fleet angle by controlling side-to-side movement of the load with tracks or guide rails. Allowing the load to move too far to one side causes stress on the drum flange which may cause damage.



1.2 Installing the Handle

1.2.1 FOR MODELS M4022PB, M4032PB, M4042PBSS, M4312PB, M4312PBSS, and M4412PB equipped with PB Brakes, install the handle as follows:

a LOOSEN THE THUMBSCREW and slide the handle toward the medallion as shown. See Figure 2A.

b LINE UP THE NOTCH in the handle with the point of the hex in the medallion. See Figure 2B.

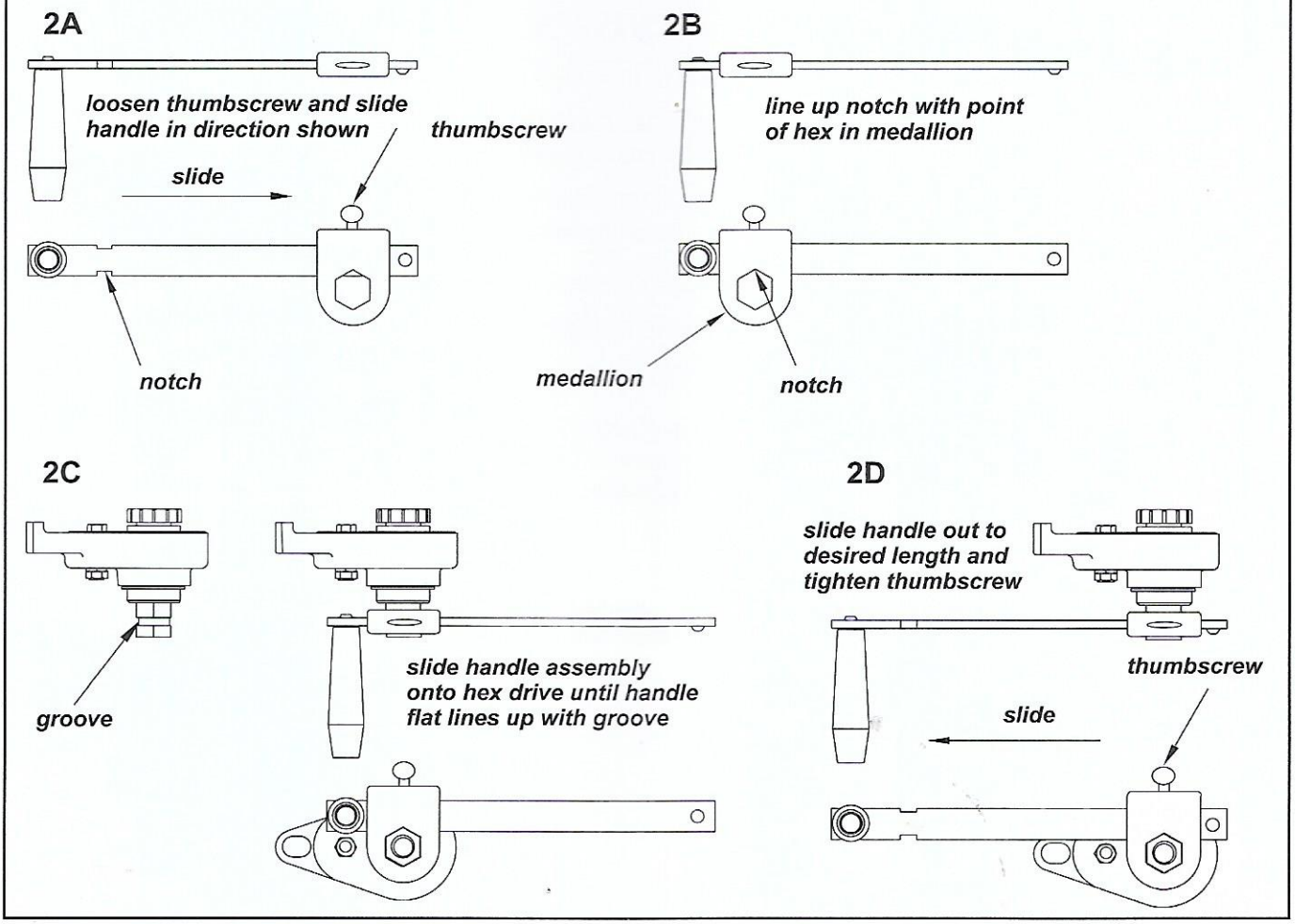
c SLIDE THE HANDLE AND MEDALLION ASSEMBLY onto the hex drive on the brake until the flat of the handle lines up with the groove in hex drive. See Figure 2C.

d SLIDE THE HANDLE outward away from the medallion to the desired length and tighten the thumbscrew. See Figure 2D.

1.2.2 FOR MODELS M4022 AND M4032, place the handle assembly on the handle shaft and secure in place with the lock washer and hex nut.

1.2.3 FOR MODELS M4312 AND M4412, place the handle assembly on the drum shaft and install the hair pin cotter in the end of the drum shaft to hold the handle in place.

Figure 2 – Installing the Handle

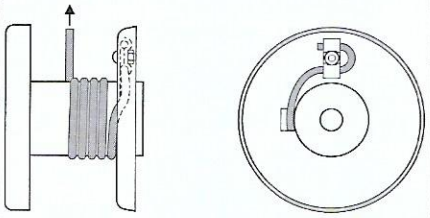


1.3 Installing the Wire Rope

Important!

- Use wire rope and other rigging equipment rated for the size of the largest load you will be moving.
- Do not drag the wire rope through dirt or debris that could cause damage, or poor operation.
- Always wear protective clothing when handling wire rope.

Figure 3A – Flange Clip Anchor



drum end view

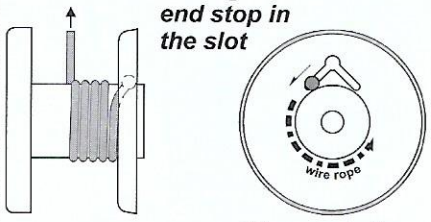
*typical installation shown right lay – underwound**

**M4042PBSS is overwound*

Wire rope assemblies can be purchased from Thern, Inc.

Figure 3B – Quick Disconnect Anchor

pull forward to lodge the end stop in the slot



drum end view

*typical installation shown right lay – underwound**

**M4042PBSS is overwound*

Wire rope assemblies can be purchased from Thern, Inc.

⚠WARNING

Install the wire rope so it is wound correctly as shown, or the winch will not work properly, and could release the load.

Install the wire rope securely to the winch drum. A poorly secured wire rope could come loose from its anchor and could release the load.

1.3.1 PURCHASE THE PROPER WIRE ROPE for your application. Keep the following in mind when selecting a wire rope. Contact a repuTable wire rope supplier for help.

a BREAKING STRENGTH of new wire rope should be at least 3 times greater than the largest load placed on the winch. If loads are lifted or pulled on an incline, the breaking strength must be at least 5 times greater than the largest load. These are minimum values and will vary with the type of load and how you are moving it.

b WIRE ROPE LAY must agree with the winding direction of the drum to help insure proper winding.

c WE RECOMMEND 7 x 19 galvanized aircraft cable for diameters up to 5/16 inch.

1.3.2 ANCHOR THE WIRE ROPE to the drum using either the flange clip or quick disconnect anchor.

a FLANGE CLIP ANCHOR. See Figure 3A.

- PASS THE END OF THE WIRE ROPE through the hole in the flange.
- INSERT THE CARRIAGE BOLT, loop the wire rope around the bolt, and install the clip so the wire rope is held by the curves of the clip.
- INSTALL THE JAM NUT and tighten it until the wire rope is flattened against the drum flange.

b QUICK DISCONNECT ANCHOR. See Figure 3B.

- PUSH THE END STOP through the large center hole in the drum flange.
- PULL THE WIRE ROPE forward to lodge the end stop in the slot.

1.3.3 TURN THE HANDLE CLOCKWISE to wind wire rope onto the drum. If wire rope unwinds from the drum when the handle is rotated clockwise, the wire rope is installed incorrectly. **Install the wire rope correctly before continuing.**

1.3.4 WIND FOUR FULL WRAPS of wire rope onto the drum by operating the winch while holding the wire rope taught. **These wraps serve as anchor wraps and must remain on the drum at all times.**

2.1 General Theory of Operation

Important!

- Limit nonuniform winding by keeping tension on the wire rope and by maintaining the proper fleet angle.
- It is your responsibility to detect and account for different factors affecting the condition and performance of the equipment.

2.1.1 THE PULL REQUIRED to move the load must not exceed the load rating of the winch. Consider the total force required to move the load, not the weight of the load.

2.1.2 THIS EQUIPMENT CAN develop forces that will exceed the load rating. It is the responsibility of the equipment user to limit the size of the load. Inspect the equipment regularly for damage according to the instructions contained in this manual.

2.1.3 USE A DISC BRAKE on all hand winches used to lift loads or pull loads on an incline.

2.1.4 PERFORMANCE RATINGS of the equipment are affected by the amount of wire rope wound on the drum, the way in which it is wound, and the way the winch is used.

a DRUM CAPACITY depends on how tightly and evenly the wire rope is wound on the drum. Actual drum capacities are usually 25-30% less than values shown in performance Tables, due to loose winding and overlapping.

b FORCE REQUIRED TO LIFT the load increases with each additional layer of wire rope wound onto the drum. The value shown in performance Tables is based on an empty drum and maximum handle length.

c LOAD RATING represents the maximum pull that can be placed on new equipment. Load ratings are assigned values for specific amounts of load travel or wire rope accumulation. The load rating decreases as layers of wire rope accumulate on the drum.

2.1.5 DUTY RATINGS refer to the type of use the equipment is subject to. Consider the following when determining duty rating.

a ENVIRONMENT: harsh environments include hot, cold, dirty, wet, corrosive, or explosive surroundings. **Protect the equipment from harsh environments when possible.**

b MAINTENANCE: poor maintenance, meaning poor cleaning, lubrication, or inspection, leads to poor operation and possible damage of the equipment. **Minimize poor maintenance by carefully following the instructions contained in this manual.**

c LOADING: severe loading includes shock loading and moving loads that exceed the load rating of the equipment. **Avoid shock loads, and do not exceed the load rating of the equipment.**

d FREQUENCY OF OPERATION: frequent or lengthy operations increase wear and shorten the life span of gears, bearings, and other components. **Increase maintenance of the equipment if used in frequent operations.**

2.1.6 AIRBORNE NOISE EMISSIONS vary depending on load and the type of winch being used. Standard configurations do not typically exceed a sound pressure level of 80dB(A) at workstations.

CONTACT THE FACTORY FOR MORE INFORMATION.

2.2 Breaking-In the Winch

2.2.1 BREAK-IN OCCURS during the first 10 hours of normal operation. During break-in, mating surfaces become polished, and clearances increase. This is desired for efficient operation of bearings and gears.

2.2.2 INSPECT THE WINCH following break-in according to the Instructions for Periodic Inspection.

2.3 Preparing for Operation

Important!

- When determining whether the load will exceed the load rating, consider the total force required to move the load.

2.3.1 CONSIDER THE OPERATION. Do not begin until you are sure you can perform the entire operation without hazard.

2.3.2 INSPECT ALL COMPONENTS of the system.

a INSPECT THE WINCH and other equipment according to the Instructions for Frequent Inspection.

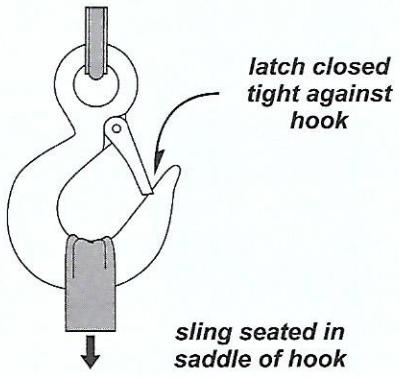
b OPERATORS must be in good health, alert, thoroughly trained in operating the equipment, and properly clothed (hard hat, safety shoes and safety glasses, no loose clothing).

c THE LOAD must be clear of other objects and free to move. Make sure the load will not tip, spin, roll away, or in any way move uncontrollably.

2.3.3 KNOW YOUR LOAD and make sure you do not exceed the load rating of the winch or any other equipment in the system.

2.4 Attaching the Load

Figure 4 – Attaching Load



⚠WARNING

Do not wrap the wire rope around the load. This damages the wire rope and could cause the load to escape. Use a sling or other approved lifting device.

2.4.1 CLEAR OBJECTS from the path of the load so you can move it freely and observe it at all times during the operation.

2.4.2 ATTACH THE LOAD using a nylon sling, or other approved lifting device. Follow the recommendations of the sling manufacturer.

a SEAT THE SLING in the saddle of the hook with the hook latch completely closed. See Figure 4.

b CENTER THE LOAD on the hook so it will remain balanced and not tip or rotate to one side.

2.5 Moving the Load

2.5.1 MOVE THE LOAD slowly and smoothly, only a small distance at first. Make sure the load is balanced and securely attached before continuing.

2.5.2 TURN THE HANDLE CLOCKWISE to wind wire rope onto the drum. If wire rope unwinds from the drum when the handle is rotated clockwise, the wire rope is installed incorrectly. **Install the wire rope correctly before continuing.**

2.5.3 OBSERVE THE WIRE ROPE as it winds onto the drum. If it becomes loose, uneven, or overlapped, stop the operation and rewind the wire rope before continuing. **Continued operation with overlapped or uneven wire rope can damage the wire rope and shorten its life.**

2.5.4 FOR MODELS M4312 AND M4412 ONLY. Move the handle to the countershaft to quickly wind or unwind UNLOADED wire rope. **Do not attempt to move a load with the handle installed on the countershaft.**

2.5.5 FOR MODELS M4022, M4032, M4312 AND M4412 ONLY. Engage the ratchet when winding wire rope onto the drum, disengage the ratchet to unwind the wire rope. **DO NOT ENGAGE THE RATCHET while the winch gears are turning.** Stop the winch before engaging the ratchet. Disengage the ratchet by moving the load slightly to remove pressure from the ratchet, then pull out and turn the ratchet knob 90 degrees. Grip the handle tightly at all times during operation. If you release the handle the load may backdrive causing the handle to spin. **Do not try to stop a spinning handle, step clear until the spinning stops.**

2.5.6 MOVE THE LOAD slowly and smoothly, only a small distance at first. Make sure the load is balanced and securely attached before continuing.

2.5.7 OBSERVE THE WIRE ROPE as it winds onto the drum. If it becomes loose, uneven, or overlapped, stop the operation and rewind the wire rope before continuing. Continued operation with overlapped or uneven wire rope can damage the wire rope and shorten its life.

Important!

- Obey a stop signal from anyone.
- Maintain tension on the wire rope to keep it tightly and evenly wound on the drum.
- If the winch and load are not visible during the entire operation, get help from another person.
- Appoint a supervisor if more than one person is involved in the operation. This will reduce confusion and increase safety.
- When lifting a load, use a tag line to keep the load from swinging or twisting, while keeping yourself away from the load.
- Remove the winch handle when the winch is not in use, to help avoid unauthorized use.

3.1 Cleaning the Winch

Clean the winch to remove dirt and help prevent rust and corrosion.

3.1.1 CLEAN THE WINCH every six months or whenever it is dirty.

a WIPE ALL EQUIPMENT to remove dirt and grease.

b LEAVE A LIGHT FILM of oil on all surfaces to protect them against rust and corrosion.

c WIPE OFF excessive amounts of oil to avoid the accumulation of dirt.

3.1.2 REMOVE ALL UNNECESSARY OBJECTS from the area surrounding the winch.

3.2 Lubricating the Winch

Important!

Increase the frequency of maintenance procedures if the winch is:

- Operated for long periods.
- Used to pull heavy loads.
- Operated in wet, dirty, hot, or cold surroundings.

⚠CAUTION

Do not over lubricate the brake bushings on models equipped with a brake. Over lubricating may cause oil to leak onto the friction discs, which may damage the friction discs or result in poor operation of the disc brake.

Lubricate the spur gears before each operation, and periodically during operation. Failure to lubricate the gears will cause damage or deformation of gear teeth.

Lubricate the winch properly to help protect it from wear and rust. Read the following instructions carefully.

3.2.1 CONSULT MANUFACTURER'S RECOMMENDATIONS for specific information on lubricating the wire rope and other equipment.

3.2.2 LUBRICATE WINCH BEARINGS AND SHAFTS at least every 6 months.

a APPLY 2 TO 3 DROPS of SAE 30 non-detergent oil to bearings and shafts at all friction points.

b ROTATE THE DRUM several times to allow the oil to penetrate, and wipe off excess oil to avoid accumulation of dirt.

Important!

- Make sure lubricant has a temperature rating appropriate for the ambient temperatures of the operation.

3.2.3 LUBRICATE WINCH GEARS before every operation and at least every 10 hours during operation.

a APPLY A LIGHT FILM of open gear lubricant to the gear teeth on all gears.

b USE SPRAYON S00201 or equivalent open gear lube. For dirty conditions use a dry lubricant such as dry graphite or Moly.

c If equipped with a guard mounted grease fitting(s) use standard grease gun to apply grease onto gears and rotate handle to distrubute until gears are evenly coated.

3.2.4 LUBRICATE THE DISC BRAKE IF APPLICABLE at least every 6 months. Place 1 or 2 drops of SAE 30 non-detergent oil into the hole in the brake housing marked "oil", and turn the brake several times to allow the oil to penetrate.

3.2.5 LUBRICATE THE WIRE ROPE and other equipment by following the manufacturer's recommendations.

3.3 Inspecting the Equipment

Important!

- Start an inspection program as soon as you put the winch into use.
- Appoint a qualified person to be responsible for regularly inspecting the equipment.
- Keep written records of inspection. This allows comparison with comments from previous inspections so you can see changes in condition or performance.

Perform frequent inspections:

- Before each operation.
- Every 3 hours during operation.
- Whenever you notice signs of damage or poor operation.

Frequent Wire Rope Inspection:

- Use ASME B30.7 as a guideline for rope inspection, replacement and maintenance.
- Check the wire rope, end connections and end fittings for corrosion kinking, bending, crushing, birdcaging or other signs of damage.
- Check the number, distribution and type of visible broken wires. See paragraph 3.3.4 b and Figure 5.
- Check the wire rope for reduction of rope diameter from loss of core support, or wear of outside wires. See Figure 7.
- Take extra care when inspecting sections of rapid deterioration such as sections in contact with saddles, sheaves, repetitive pickup points, crossover points and end connections.

⚠WARNING

Do not use damaged or malfunctioning equipment. Place an "OUT OF ORDER" sign on the winch. Do not use the winch until the sign is removed by a qualified maintenance person who has completely corrected the problem.

Inspect the winch to detect signs of damage or poor operation before they become hazardous.

3.3.1 CONSULT APPLICABLE CODES AND REGULATIONS for specific rules on inspecting the winch and other equipment.

3.3.2 CONSULT MANUFACTURER'S RECOMMENDATIONS for information on inspecting the wire rope and other equipment.

3.3.3 **Instructions for Frequent Inspection**

a VISUALLY INSPECT the entire winch and all other equipment involved in the operation.

- Check all equipment for cracks, dents, bending, rust, wear, corrosion and other damage.
- Make sure the wire rope is installed correctly and anchored securely to the drum.
- Make sure the winch and brake are properly lubricated.
- Make sure fasteners are installed to securely hold the handle in place.
- Make sure mounting fasteners are tightened securely.
- Make sure the foundation is in good condition, and capable of supporting the winch and its load under all load conditions.

b TEST WINCH PERFORMANCE by operating the winch with a load not exceeding the load rating.

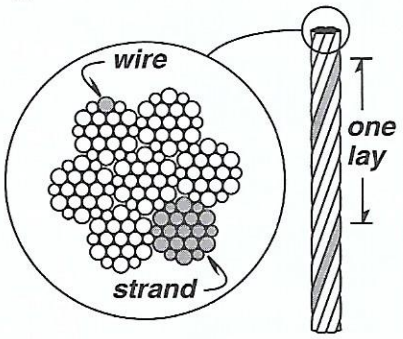
- Listen for unusual noises, and look for signs of damage as you operate the winch.
- Make sure the wire rope winds evenly and tightly onto the drum. If it is loose or uneven, rewind it before continuing.
- Make sure the handle rotates freely in both directions.
- Make sure the load moves smoothly without hesitation or strain.
- On models equipped with a ratchet, make sure the ratchet engages and disengages completely.
- On models equipped with a brake, make sure the disc brake ratchet pawl clicks firmly as the brake handle is turned clockwise.
- On models equipped with a brake, check the brake. Raise the load, then lower it and stop it a few feet off the ground. If the load continues to coast or creep under normal operating conditions, the friction discs may be worn and in need of replacement. Contact the factory.

Completely correct all problems before continuing. Use the Troubleshooting Chart to help determine the cause of certain problems. See Table 2.

Perform periodic inspections:

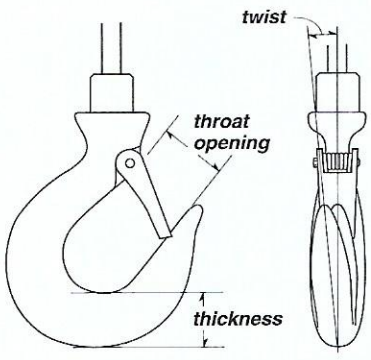
- Every 6 months.
- Whenever you return the winch to service from storage.
- Whenever you notice damage or poor operation in a frequent inspection.
- Whenever you have, or think you may have, overloaded or shock loaded the winch.

Figure 5 – Broken Wires



Wire rope assembly must be replaced if more than 6 wires are broken in one lay, or if more than 3 wires are broken in one strand in one lay.

Figure 6 – Load Hook Inspection



The wire rope assembly must be replaced if the throat opening is 15% wider than nominal, if the thickness is 10% less than nominal, or if the hook is twisted 10° or more.

3.3.4 **Instructions for Periodic Inspection, see Table 1.**

a VISUALLY INSPECT the winch and all other equipment.

- Disassembly may be required in order to properly inspect individual components. Contact factory for assembly/disassembly instructions. Disassembly of the winch or brake before contacting Thern, Inc. voids all warranties.
- Check the finish for wear, flaking, or other damage.
- Check all equipment, including wire rope for cracks, dents, bending, rust, wear, corrosion and other damage. If the winch was overloaded, or if you notice cracks and other signs of overloading and damage promptly remove equipment from use and have it repaired or replaced. **DO NOT CONTINUE TO USE DAMAGED OR OVERLOADED EQUIPMENT OR WIRE ROPE.**
- Check all fasteners for stripped threads, wear, bending, and other damage.
- Check the foundation for cracks, corrosion, and other damage.
- Make sure the winch and brake are properly lubricated.
- Make sure all labels and plates are readable, firmly attached, free of damage and clean. Replacements are available from the factory.

b REMOVE THE WIRE ROPE entirely from the drum.

- Always wear protective clothing when handling wire rope.
- Check the entire length of wire rope for bent wires, crushed areas, broken or cut wires, corrosion, and other damage. Carefully inspect areas that pass over sheaves or through roller guides.
- Note the location and concentration of broken wires. Replace wire rope if more than 6 wires are broken in one lay, or more than 3 wires are broken in one strand in one lay. See Figure 5.
- Make sure the load hook or other device is securely attached to the wire rope, and the wire rope where it is attached is not frayed, corroded, broken, or otherwise damaged.
- Measure the throat opening, thickness, and twist of the hook. Replace the hook if it shows signs of damage. See Figure 6.
- Make sure hook latch opens without binding and closes when released.
- Check the anchor holes in the drum flange for signs of wear or distortion.

c PLACE enough weight to keep the wire rope straight and tightly drawn.

- Measure the diameter of the wire rope, especially in areas where wear is noticeable. Replace the wire rope if the diameter measures below the minimum diameter at any point. See Figure 7.

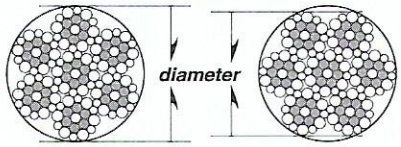
d REMOVE THE WINCH from the foundation.

- Check fasteners for stripped threads, wear, bends, and other damage.
- Check the frame for bending, distortion, cracks and other damage. A bent frame is caused by overloading, and is a sign that your application may require a winch with a larger load rating.

e MOVE THE DRUM with your hands.

- Check for excessive movement indicating worn or loose gears, bearings, or shafts.

Figure 7 – Rope Diameter



The wire rope assembly must be replaced if the diameter measures less than the minimum diameter at any point.

wire rope diameter	minimum diameter
1/8 in	7/64 in (.1094 in)
3/16 in	11/64 in (.1719 in)
1/4 in	15/64 in (.2344 in)
5/16 in	19/64 in (.2969 in)

f FASTEN THE WINCH securely to the foundation.

g INSTALL THE WIRE ROPE.

h TEST WINCH PERFORMANCE by operating the winch with a load equal to the load rating.

- Listen for unusual noises, and look for signs of damage as you operate the winch.
- Make sure the wire rope winds evenly and tightly onto the drum. If it is loose or uneven, rewind it before continuing.
- Observe the rotating drum, look for signs of loose or misaligned bearings.
- Make sure the handle rotates freely in both directions.
- Make sure the load moves smoothly without hesitation or strain.
- On models equipped with a ratchet, make sure the ratchet engages and disengages completely.
- On models equipped with a brake, make sure the disc brake ratchet pawl clicks firmly as the brake handle is turned clockwise.
- On models equipped with a brake, check the brake. Raise the load, then lower it and stop it a few feet off the ground. If the load continues to coast or creep under normal operating conditions, the friction discs may be worn and in need of replacement. Contact the factory.

Completely correct all problems before continuing. Use the troubleshooting chart to help determine the cause of certain problems. See Table 2.

Table 1 – Inspection Checklist

checked boxes indicate damage or problem in need of repair

	damages	problems
general	☐ finish weathered, flaking, otherwise damaged	☐ winch jerks or hesitates during operation
	☐ parts cracked, bent, rusted, worn, otherwise damaged	☐ unusual noises, other signs of malfunction
fasteners	☐ stripped threads, bent, worn, otherwise damaged	☐ loose, not tightened to proper torque
gears	☐ excessively worn, cracked, corroded, otherwise damaged	☐ loose or improperly lubricated
ratchet	☐ ratchet corroded, cracked, worn, otherwise damaged	☐ ratchet does not engage or disengage
brake assembly	☐ brake corroded, cracked, worn, otherwise damaged	☐ brake does not operate properly
drum	☐ anchor hole worn, distorted, otherwise damaged	☐ excessive movement or backlash
wire rope	☐ bent, crushed, otherwise damaged	☐ wire rope loosely or unevenly wound
	☐ broken wires, see Figure 5	
	replace if more than 6 wires in one lay,	number per strand =
	or 3 wires in one strand in one lay, are broken	number per lay =
	☐ diameter reduced, see Figure 7	
	replace if diameter is excessively worn	diameter =
end connections	☐ corroded, rusted, worn, otherwise damaged	☐ not securely attached
load hook	☐ twisted, bent, worn, otherwise damaged, see Figure 6	☐ hook latch fails to close when released
	replace if twist is 10 degrees or more	twist =
	replace if throat width is 15% larger than nominal	throat width =
	replace if thickness is 10% less than nominal	thickness =
labels and plates	☐ dirty, illegible, otherwise damaged	☐ loosely attached or missing

comments:

authorized signature: _____ **date** _____

3.4 Repairing the Winch

Important!

- It is your responsibility to determine when to replace parts. When considering whether to continue using a part or to replace it, remember that replacing it is the best way to avoid further equipment damage.
- Replace all spring pins and retaining rings when you disassemble the winch or brake for repair or replacement.
- Appoint a qualified person to be responsible for all repairs to the equipment.

3.4.1 GET FACTORY AUTHORIZATION for all repairs. Unauthorized repairs will void the warranty, and may lead to damage or failure of the winch.

3.4.2 REPLACE DAMAGED OR POORLY OPERATING PARTS with Thern repair parts.

3.4.3 REFINISH AREAS where the paint is worn or flaking. A good finish helps to protect against corrosion and weather damage.

a REMOVE THE FINISH from damaged areas, down to the bare metal.

b CLEAN THE AREA thoroughly.

c REPAINT with a high quality primer and finishing coat.

3.4.4 TO ORDER REPAIR PARTS, contact your local dealer. Include the following information when ordering:

- model number
- serial number (or code number)
- part number
- date purchased, and from whom
- description of what happened, or what is wrong
- your name and return address

4.1 Transporting the Winch

Important!

- Keep a record of what you ship, and when you send it.

4.1.1 PACK THE WINCH using the original packaging materials, if possible.

4.1.2 SEAL THE WINCH in plastic with a desiccant to help protect it from rust, corrosion, and other damage.

4.1.3 INSPECT THE WINCH according to the Instructions for Periodic Inspection before installing it in a new location.

4.2 Storing the Winch

4.2.1 SEAL THE WINCH in plastic with a desiccant to help protect it from rust, corrosion, and other damage.

4.2.2 STORE THE WINCH in a cool clean place away from corrosive chemicals and moisture.

4.2.3 ROTATE THE DRUM periodically to keep bearing and gear surfaces from becoming lacquered.

4.2.4 INSPECT THE WINCH according to the Instructions for Periodic Inspection before installing it for operation.

Table 2 – Troubleshooting Chart

Contact the factory for detailed instructions if you are required to disassemble the winch or brake for any reason. Disassembly of the winch or brake before contacting Thern, Inc. voids all warranties.

problem	cause	correction
handle turns, drum doesn't turn	• loose or broken spring pins	inspect winch and brake, repair as necessary
	• loose, stripped or broken gears	inspect gears and repair as necessary
handle turns hard or not at all	• ratchet on winch engaged	disengage winch ratchet
	• load too heavy	lighten load
	• spring pins loose or broken on winch or brake	inspect winch and brake, repair as necessary
	• disc brake damaged or locked	inspect brake, repair as necessary
	• gears or bearings broken or locked	inspect and repair as necessary
brake does not operate properly	• friction discs worn or damaged	inspect and replace as necessary
	• friction discs damaged from over lubrication	inspect and replace as necessary
	• disc brake ratchet pawl damaged	inspect and repair as necessary
excessively worn gears or bearings	• load too heavy	lighten load
(excessive backlash)	• poor lubrication of gears or bearings	inspect and relubricate as necessary
overheating	• operated too long without rest	allow to cool
	• load too heavy	lighten load
	• poor lubrication	inspect and lubricate as necessary
	• bearing seized up	inspect and replace as necessary
unusual noises		
high pitched squeak	• poor lubrication	inspect and relubricate as necessary
grinding noise	• contaminated lubrication	clean and relubricate winch
	• dirt in brake or winch gears	inspect and clean as necessary
	• broken gears or bearings	inspect and replace as necessary
rattling noise	• loose bolts, set screws or other fasteners	tighten all bolts and other fasteners
uneven clicking noise in brake	• broken gear tooth in brake	inspect and repair as necessary
weak clicking noise in brake	• spring or ratchet pawl dirty or damaged	inspect and clean or repair as necessary
	• worn brake ratchet pawl, gear, or spring	inspect and replace as necessary
no clicking noise in brake	• ratchet incorrectly installed	disassemble and install correctly
	• ratchet pawl damaged or worn excessively	inspect and replace as necessary

Spur Gear Hand Winch Performance Characteristics [1] Imperal

model number	description	load rating (lb) 1st layer	mid drum	full drum	wire rope dia. (in)	drum capacity (ft)[2] 1st layer	mid drum	full drum	single gear ratio	double gear ratio	force[3] to lift 1000 lb	approx. ship wt. (lb)
M4022	1000 lb – marine duty (for pulling only)	1000	700	500	1/8 3/16	4 3	52 26	130 57	2.85:1	—	40 lb	12
M4022PB	1000 lb – marine duty with brake (for lifting)	1000	700	500	1/8 3/16	4 3	52 26	130 57	2.85:1	—	41 lb	17
M4032	1000 lb – marine duty (for pulling only)	1000	700	500	1/8 3/16	11 7	110 51	250 110	2.85:1	—	40 lb	14
M4032PB	1000 lb – marine duty with brake (for lifting)	1000	700	500	1/8 3/16	11 7	110 51	250 110	2.85:1	—	41 lb	18
M4042PBSS	1000 lb – stainless steel with brake (for lifting)	1000	800	600	1/8 3/16 1/4	12 8 5	110 48 27	240 110 59	3.83:1	—	46 lb (single gear)	24
M4312	2000 lb – marine duty (for pulling only)	2000	1600	1200	3/16 1/4 5/16	8 5 4	48 27 17	110 59 39	3.83:1	14.7:1	20 lb (double gear)	23
M4312PB	2000 lb – marine duty with brake (for lifting)	2000	1600	1200	3/16 1/4 5/16	8 5 4	48 27 17	110 59 39	—	14.7:1	17 lb (double gear)	28
M4312PBSS	2000 lb – stainless steel with brake (for lifting)	2000	1600	1200	3/16 1/4 5/16	8 5 4	48 27 17	110 59 39	—	14.7:1	17 lb (double gear)	28
M4412	2000 lb – marine duty (for pulling only)	2000	1600	1200	3/16 1/4 5/16	18 14 11	97 52 35	210 120 77	3.83:1	14.7:1	20 lb (double gear)	25
M4412PB	2000 lb – marine duty with brake (for lifting)	2000	1600	1200	3/16 1/4 5/16	18 14 11	97 52 35	210 120 77	—	14.7:1	17 lb (double gear)	30

[1] Performance Characteristics are for standard products referred to in this manual. Non-standard products may vary from the original design. Contact Thern, Inc. for this information.
[2] Actual drum capacities may be 25-30% less, due to nonuniform winding. Wire rope tension will also affect drum capacity.
[3] Approximate handle force required to lift 1000 lb with an empty drum, and maximum handle length.

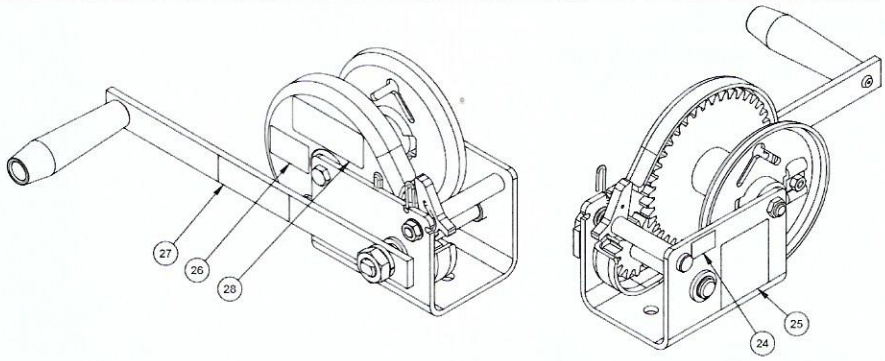
Spur Gear Hand Winch Performance Characteristics [1] Metric

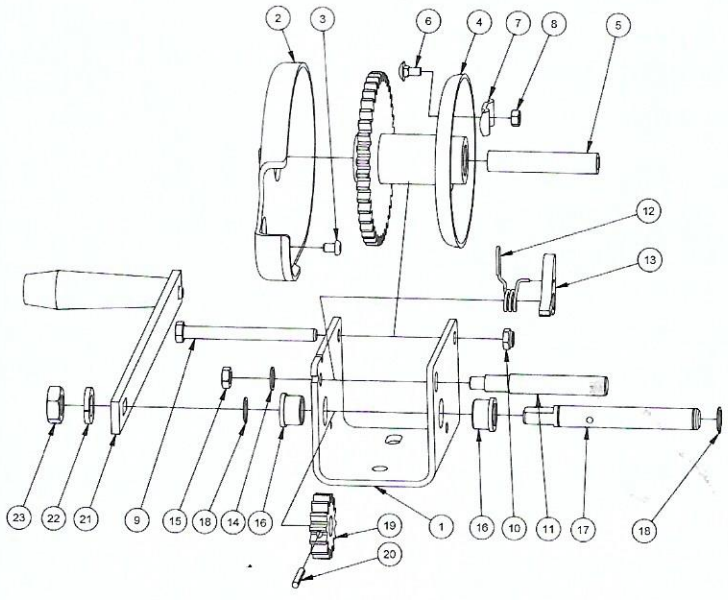
model number	description	load rating (kg) 1st layer	mid drum	full drum	wire rope dia. (mm)	drum capacity (m)[2] 1st layer	mid drum	full drum	single gear ratio	double gear ratio	force[3] to lift 450 kg	approx. ship wt. (kg)
M4022	450 kg – marine duty (for pulling only)	450	315	225	3 5	1.2 0.9	52 26	15.8 7.9	2.85:1	—	19	6
M4022PB	450 kg – marine duty with brake (for lifting)	450	315	225	3 5	1.2 0.9	52 26	15.8 7.9	2.85:1	—	19	8
M4032	450 kg – marine duty (for pulling only)	450	315	225	3 5	3.3 2.1	110 51	33.5 15.5	2.85:1	—	19	7
M4032PB	1450 kg – marine duty with brake (for lifting)	450	315	225	3 5	3.3 2.1	110 51	33.5 15.5	2.85:1	—	19	9
M4042PBSS	450 kg – stainless steel with brake (for lifting)	450	360	270	3 5 6	3.6 2.4 1.5	110 48 27	33.5 14.6 8.2	3.83:1	—	21 (single gear)	10
M4312	905 kg – marine duty (for pulling only)	905	725	540	5 6 8	2.4 1.5 1.2	48 27 17	14.6 8.2 5.1	3.83:1	14.7:1	10 (double gear)	11
M4312PB	905 kg – marine duty with brake (for lifting)	905	725	540	5 6 8	2.4 1.5 1.2	48 27 17	14.6 8.2 5.1	—	14.7:1	8 (double gear)	13
M4312PBSS	905 kg – stainless steel with brake (for lifting)	905	725	540	5 6 8	2.4 1.5 1.2	48 27 17	14.6 8.2 5.1	—	14.7:1	8 (double gear)	13
M4412	905 kg – marine duty (for pulling only)	905	725	540	5 6 8	5.4 4.2 3.3	97 52 35	29.5 15.8 10.6	3.83:1	14.7:1	10 (double gear)	12
M4412PB	905 kg– marine duty with brake (for lifting)	905	725	540	5 6 8	5.4 4.2 3.3	97 52 35	29.5 15.8 10.6	—	14.7:1	8 (double gear)	14

[1] Performance Characteristics are for standard products referred to in this manual. Non-standard products may vary from the original design. Contact Thern, Inc. for this information.
[2] Actual drum capacities may be 25-30% less, due to nonuniform winding. Wire rope tension will also affect drum capacity.
[3] Approximate handle force required to lift 450 kg with an empty drum, and maximum handle length.

Spur Gear Hand Winch Model M4022

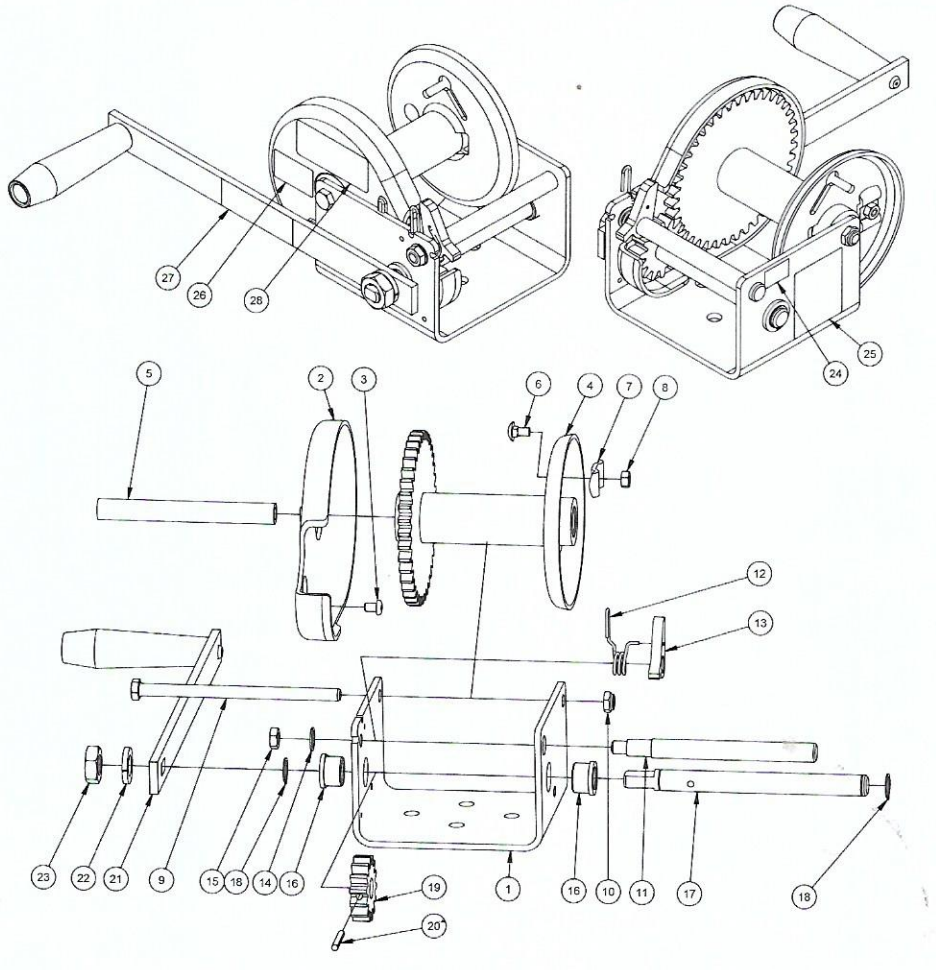
ITEM	PART NUMBER	QTY	DESCRIPTION	MASS
1	A25587	1	FRAME	2.64
2	A25558	1	DRUM GEAR GUARD	.20
3	A14146	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
4	A25588	1	DRUM ASSEMBLY	3.87
5	A25591	1	SPACER DRUM .625DIA X 11GA X 3.235	.18
6	A3333	1	CARIGBOLT .250-20NC X .500 SST	.01
7	A5922	1	TIE DOWN	.04
8	A3331	1	NUT HEX .250-20NC SST 18-8	.01
9	A3492	1	CAPSCR HEXHD .375-16NC X 4.000 ZNPL GR5	.14
10	A11913	1	NUT HEX JAM NYLK .375-16NC SST 18-8	.01
11	A25676	1	RATCHET PIN	.26
12	B1145	1	RATCHET SPRING	.01
13	A25680	1	RATCHET PAWL	.08
14	A3939	1	WASHER SPLK INT .375 X .692 X .035 SST	.00
15	A3330	1	NUT HEX JAM .375-16NC SST 18-8	.01
16	A1003	2	FLANGE BEARING	.07
17	A25677	1	HANDLE SHAFT	.49
18	A4136	2	RING RET EXT .625 SST	.00
19	A25593	1	PINION STEEL SPUR M4021	.29
20	A4851	1	PIN GROOVE TYPE A .187 X 1.375 ALYSTL	.01
21	A25678	1	HANDLE ASSEMBLY	1.26
22	A3327	1	WASHER HELSPRLK .625 X 1.086 X .156 SST	.03
23	A3329	1	NUT HEX JAM .625-11NC SST 316	.07
24	10477	1	LABEL DATE OF MANUFACTURE	.00
25	A1978	1	LABEL WARNING	.00
26	A2175	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
27	A2691	1	LABEL WARNING NO AUTOMATIC BRAKE	.00
28	A25761	1	LABEL MODEL/CAPACITY M4022	.00





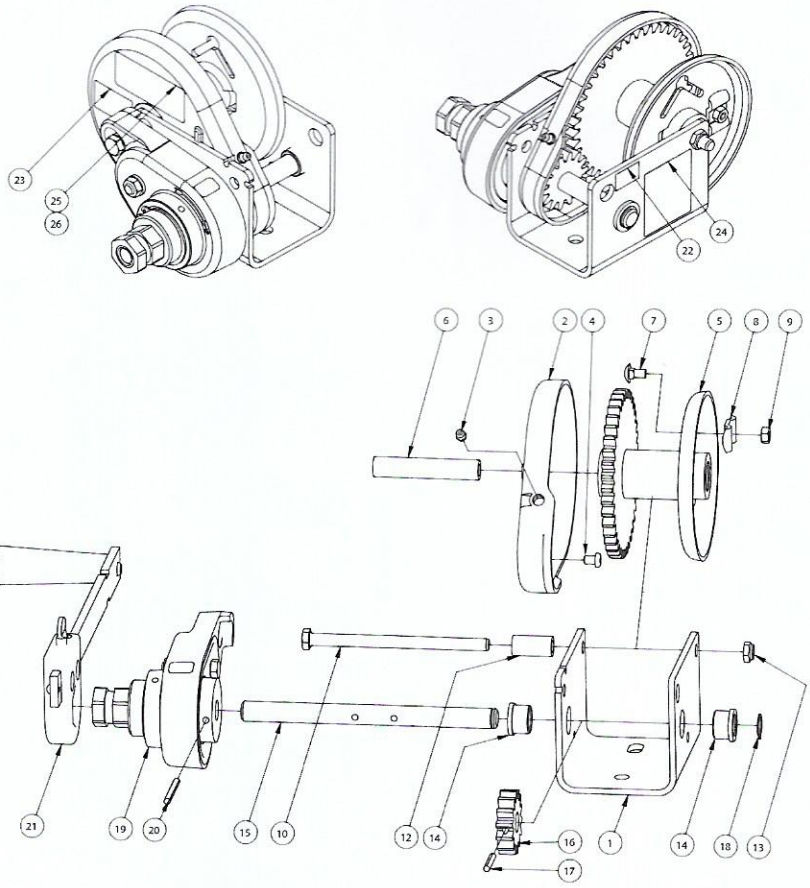
Spur Gear Hand Winch — Model M4032

ITEM	PART NUMBER	QTY	DESCRIPTION	MASS
1	A25666	1	FRAME	3.17
2	A25558	1	DRUM GEAR GUARD	.20
3	A14146	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
4	A25668	1	DRUM ASSEMBLY	4.58
5	A25671	1	SPACER DRUM .625DIA X 11GA X 5.30	.29
6	A3333	1	CARIGBOLT .250-20NC X .500 SST	.01
7	A5922	1	TIE DOWN	.04
8	A3331	1	NUT HEX .250-20NC SST 18-8	.01
9	A3520	1	CAPSCR HEXHD .375-16NC X 6.000 ZNPL GR5	.21
10	A11913	2	NUT HEX JAM NYLK .375-16NC SST 18-8	.01
11	A25686	1	RATCHET PIN	.41
12	B1145	1	RATCHET SPRING	.01
13	A25680	1	RATCHET PAWL	.08
14	A3939	1	WASHER SPLK INT .375 X .692 X .035 SST	.00
15	A3330	1	NUT HEX JAM .375-16NC SST 18-8	.01
16	A1003	2	FLANGE BEARING	.07
17	A25687	1	HANDLE SHAFT	.60
18	A4136	2	RING RET EXT .625 SST	.00
19	A25593	1	PINION STEEL SPUR M4021	.29
20	A4851	1	PIN GROOVE TYPE A .187 X 1.375 ALYSTL	.01
21	A25678	1	HANDLE ASSEMBLY	1.26
22	A3327	1	WASHER HELSPRLK .625 X 1.086 X .156 SST	.03
23	A3329	1	NUT HEX JAM .625-11NC SST 316	.07
24	10477	1	LABEL DATE OF MANUFACTURE	.00
25	A1978	1	LABEL WARNING	.00
26	A2175	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
27	A2691	1	LABEL WARNING NO AUTOMATIC BRAKE	.00
28	A25765	1	LABEL MODEL/CAPACITY M4032	.00



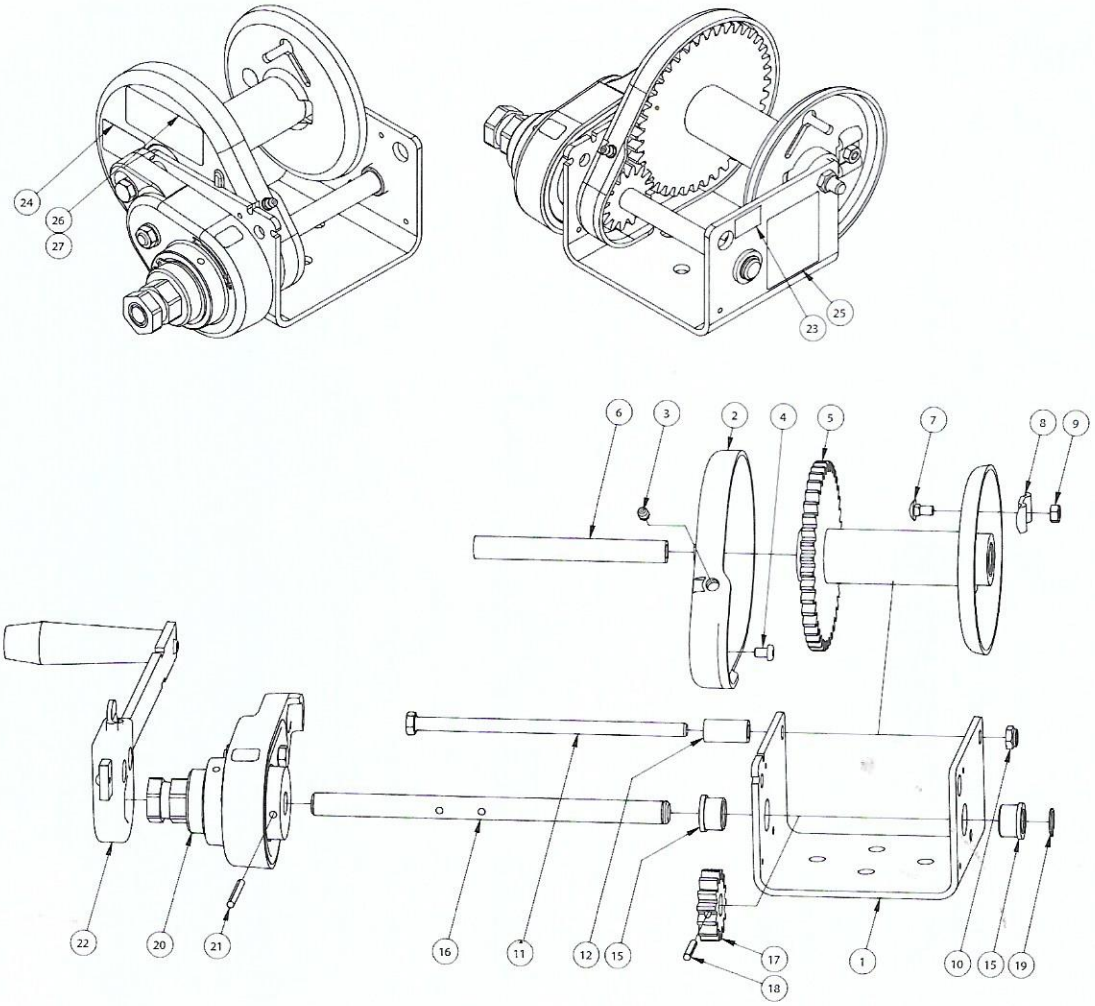
Spur Gear Hand Winch — Model M4022PB

ITEM	PART NUMBER	QTY (4022PB)	QTY (4022PB-K)	DESCRIPTION	MASS
1	A25587	1	1	FRAME	2.64
2	A25506	1	1	DRUM GEAR GUARD	.22
3	A25661	1	1	ZERK .25-28 UNF SST	.01
4	A14146	1	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
5	A25588	1	1	DRUM ASSEMBLY	3.87
6	A25591	1	1	SPACER DRUM .625DIA X 11GA X 3.235	.18
7	A3333	1	1	CARIGBOLT .250-20NC X .500 SST	.01
8	A5922	1	1	TIE DOWN	.04
9	A3331	1	1	NUT HEX .250-20NC SST 18-8	.01
10	A3519	1	1	CAPSCR HEXHD .375-16NC X 5.500 ZNPL GR5	.19
12	A25603	1	1	SPACER .391ID X .56OD X 1.212	.07
13	A11913	1	1	NUT HEX JAM NYLK .375-16NC SST 18-8	.01
14	A1003	2	2	FLANGE BEARING	.07
15	A25592	1	1	INPUT SHAFT	.66
16	A25593	1	1	PINION STEEL SPUR M4021	.29
17	A22352	1	1	PIN GROOVE TYPE A .187 X 1.250 SST	.01
18	A4136	1	1	RING RET EXT .625 SST	.00
19	A25594	1	1	DISC BRAKE ASSEMBLY	3.96
20	A22351	1	1	PIN GROOVE TYPE A .187 X 2.000 SST	.02
21	A25604	1	1	HANDLE MEDALLION ASSEMBLY	2.32
22	10477	1	1	LABEL DATE OF MANUFACTURE	.00
23	A2175	1	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
24	A1978	1	1	LABEL WARNING	.00
25	A25757	1	0	LABEL MODEL/CAPACITY CE M4022PB	.00
26	A25758	0	1	LABEL MODEL/CAPACITY CE M4022PB-K	.00



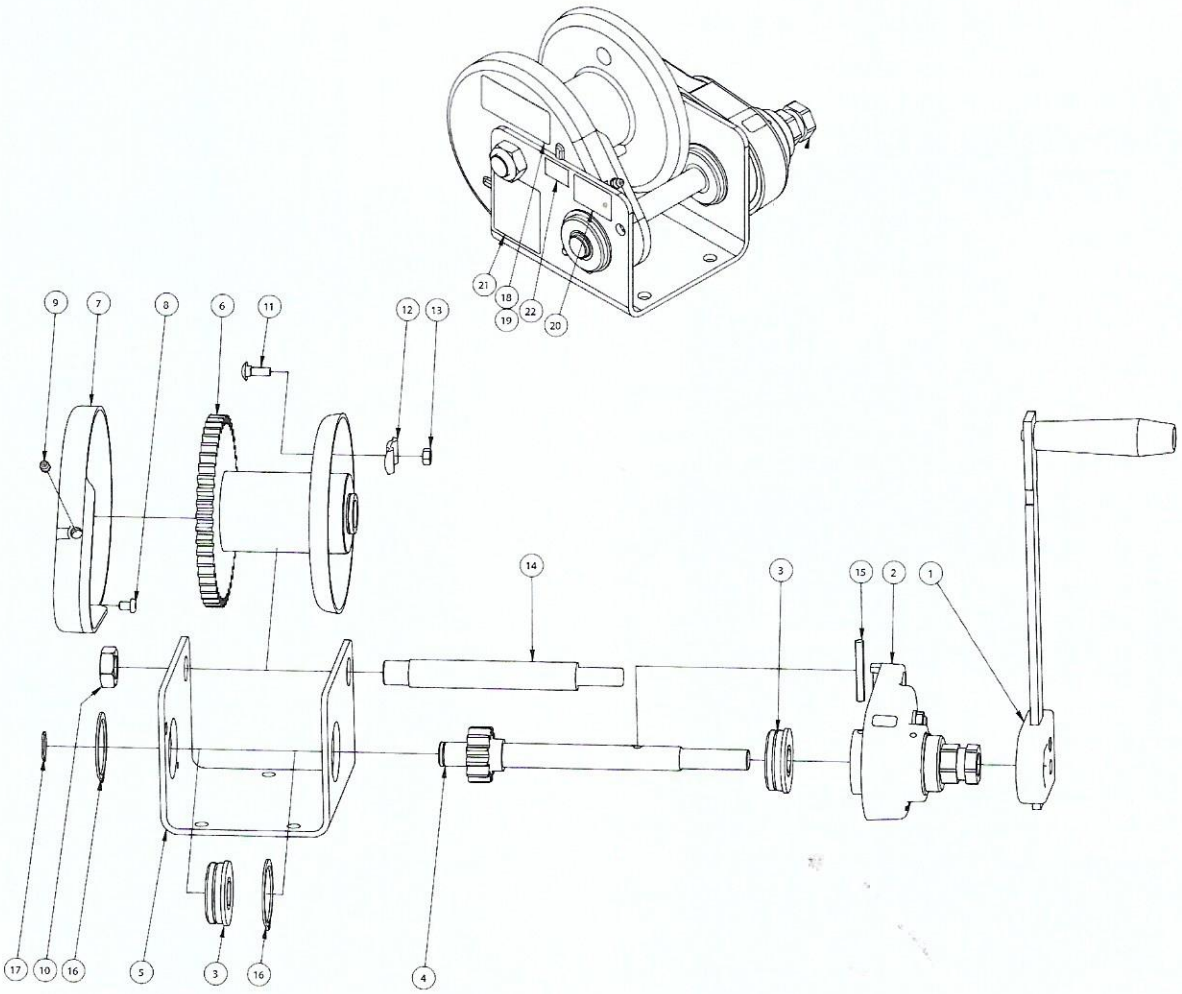
Spur Gear Hand Winch Model M4032PB

ITEM	PART NUMBER	QTY (4032PB)	QTY (4032PB-K)	DESCRIPTION	MASS
1	A25666	1	1	FRAME	3.16
2	A25506	1	1	DRUM GEAR GUARD	.22
3	A25661	1	1	ZERK .25-28 UNF SST	.01
4	A14146	1	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
5	A25668	1	1	DRUM ASSEMBLY	4.58
6	A25671	1	1	SPACER DRUM .625DIA X 11GA X 5.30	.29
7	A3333	1	1	CARIGBOLT .250-20NC X .500 SST	.01
8	A5922	1	1	TIE DOWN	.04
9	A3331	1	1	NUT HEX .250-20NC SST 18-8	.01
10	A11913	1	1	NUT HEX JAM NYLK .375-16NC SST 18-8	.01
11	A3528	1	1	CAPSCR HEXHD .375-16NC X 7.500 ZNPL GR5	.25
12	A25603	1	1	SPACER .391ID X .56OD X 1.212	.07
15	A1003	2	2	FLANGE BEARING	.07
16	A25667	1	1	INPUT SHAFT	.83
17	A25593	1	1	PINION STEEL SPUR M4021	.29
18	A22352	1	1	PIN GROOVE TYPE A .187 X 1.250 SST	.01
19	A4136	1	1	RING RET EXT .625 SST	.00
20	A25594	1	1	DISC BRAKE ASSEMBLY	3.96
21	A22351	1	1	PIN GROOVE TYPE A .187 X 2.000 SST	.02
22	A25604	1	1	HANDLE MEDALLION ASSEMBLY	2.32
23	10477	1	1	LABEL DATE OF MANUFACTURE	.00
24	A2175	1	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
25	A1978	1	1	LABEL WARNING	.00
26	A25763	1	0	LABEL MODEL/CAPACITY CE M4032PB	.00
27	A25764	0	1	LABEL MODEL/CAPACITY CE M4032PB-K	.00



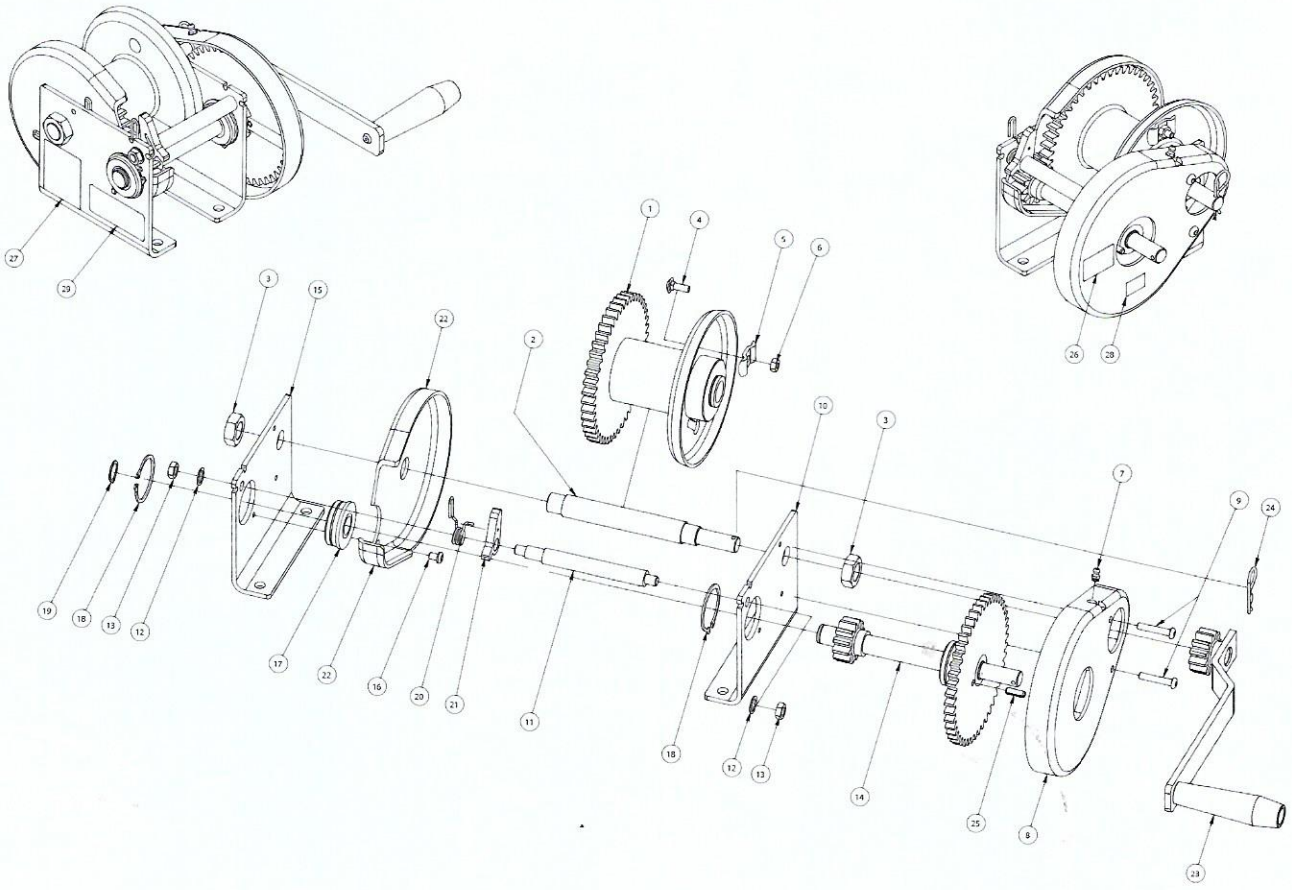
Spur Gear Hand Winch — Model M4042PBSS

ITEM	PART NUMBER	QTY (4042PBSS)	QTY (4042PBSS-K)	DESCRIPTION	MASS
1	B4217	1	1	HANDLE MEDALLION ASSEMBLY	2.42
2	C3825	1	1	DISC BRAKE ASSEMBLY	4.27
3	A4420	2	2	BEARING HOUSING ASSEMBLY	.43
4	B2415	1	1	COUNTER SHAFT WELDMENT	1.44
5	A25579	1	1	FRAME	4.08
6	A25575	1	1	DRUM ASSEMBLY	7.50
7	A25509	1	1	DRUM GEAR GUARD	.26
8	A14146	1	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
9	A25661	1	1	ZERK .25-28 UNF SST	.01
10	A3886	1	1	NUT HEX JAM .750-16NF SST 18-8	.09
11	A2846	1	1	CARIGBOLT .250-20NC X .750 SST	.02
12	A5922	1	1	TIE DOWN	.04
13	A3331	1	1	NUT HEX .250-20NC SST 18-8	.01
14	B2410	1	1	DRUM SHAFT	1.10
15	A4946	1	1	PIN GROOVE TYPE A .250 X 2.000 SST	.03
16	A4409	2	2	RING RET EXT 1.875 SST	.02
17	A4408	1	1	RING RET EXT .750 SST	.00
18	A25772	1	0	LABEL MODEL/CAPACITY CE M4042PBSS	.00
19	A25773	0	1	LABEL MODEL/CAPACITY CE M4042PBSS-K	.00
20	A2175	1	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
21	A1978	1	1	LABEL WARNING	.00
22	10477	1	1	LABEL DATE OF MANUFACTURE	.00



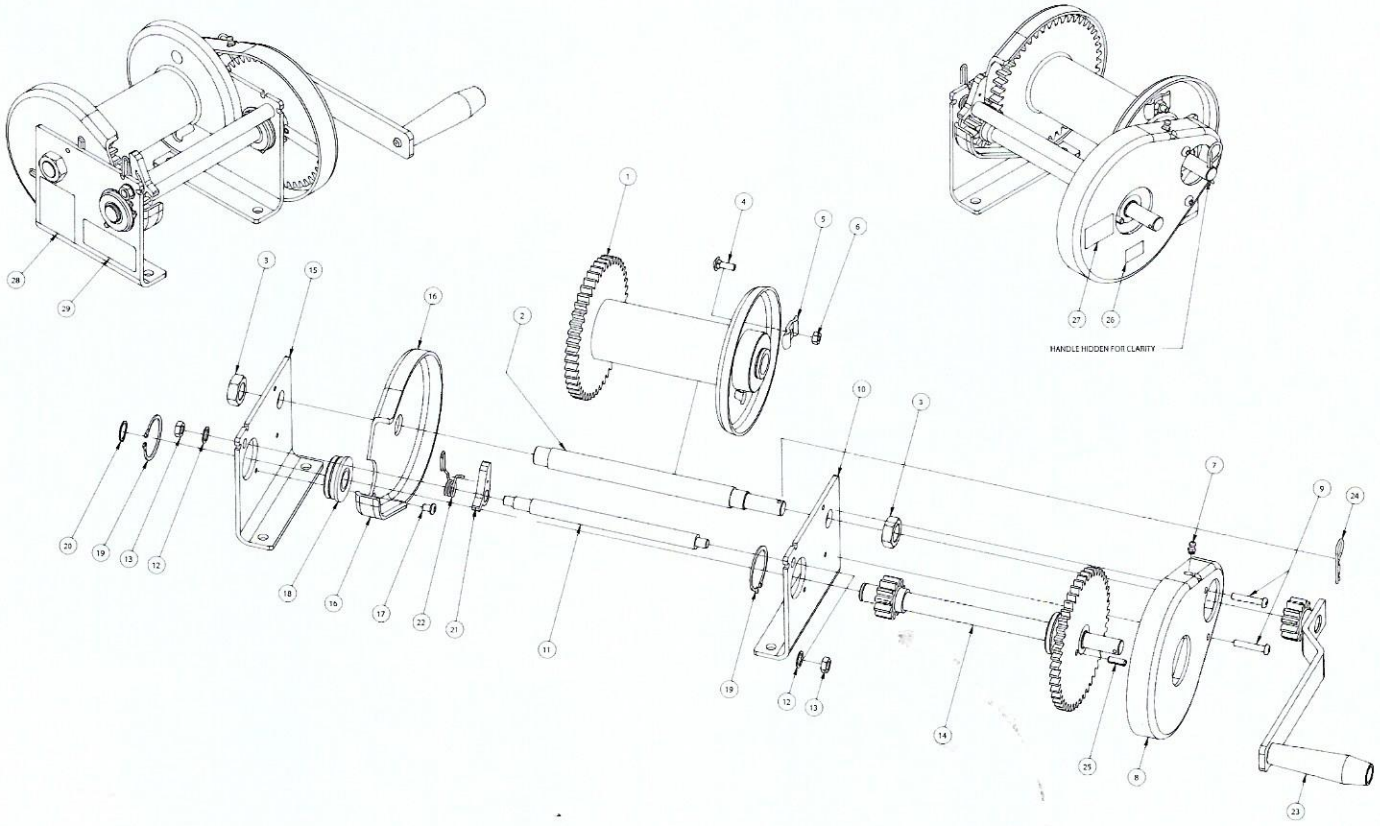
Spur Gear Hand Winch — Model M4312

ITEM	PART NUMBER	QTY	DESCRIPTION	MASS
1	A25616	1	DRUM ASSEMBLY	7.34
2	A25703	1	SHAFT DRUM M4312PB X 3.00 LG DRUM NORLS	1.10
3	A3886	2	NUT HEX JAM .750-16NF SST 18-8	.09
4	A2846	1	CARIGBOLT .250-20NC X .750 SST	.02
5	A5922	1	TIE DOWN	.04
6	A3331	1	NUT HEX .250-20NC SST 18-8	.01
7	A25661	1	ZERK .25-28 UNF SST	.01
8	A25508	1	COUNTER SHAFT GEAR GUARD	.30
9	A25557	2	CAPSCR BTNHD .250-20NC X 1.375 SST 18-8	.02
10	A25614	1	RIGHT SIDE FRAME PLATE	1.91
11	A25627	1	FRAME ROD M4312, M4312PB	.43
12	A3939	2	WASHER SPLK INT .375 X .692 X .035 SST	.00
13	A3330	2	NUT HEX JAM .375-16NC SST 18-8	.01
14	A25626	1	COUNTER SHAFT ASSEMBLY	4.49
15	A25615	1	LEFT SIDE FRAME PLATE	1.91
16	A14146	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
17	A25620	1	BEARING HOLDER ASSEMBLY	.36
18	A3883	2	RING RET EXT 1.750 SST	.01
19	A4408	1	RING RET EXT .750 SST	.00
20	B1145	1	RATCHET SPRING	.01
21	A25680	1	RATCHET PAWL	.08
22	A25559	1	DRUM GEAR GUARD	.24
23	A25704	1	HANDLE ASSEMBLY	1.66
24	A4665	1	PIN COTTER HAIR	.01
25	A4290	1	PIN SLOTTED SPRING .250 X .750 SST	.01
26	A2175	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
27	A1978	1	LABEL WARNING	.00
28	10477	1	LABEL DATE OF MANUFACTURE	.00
29	A25766	1	LABEL MODEL/CAPACITY M4312	.00



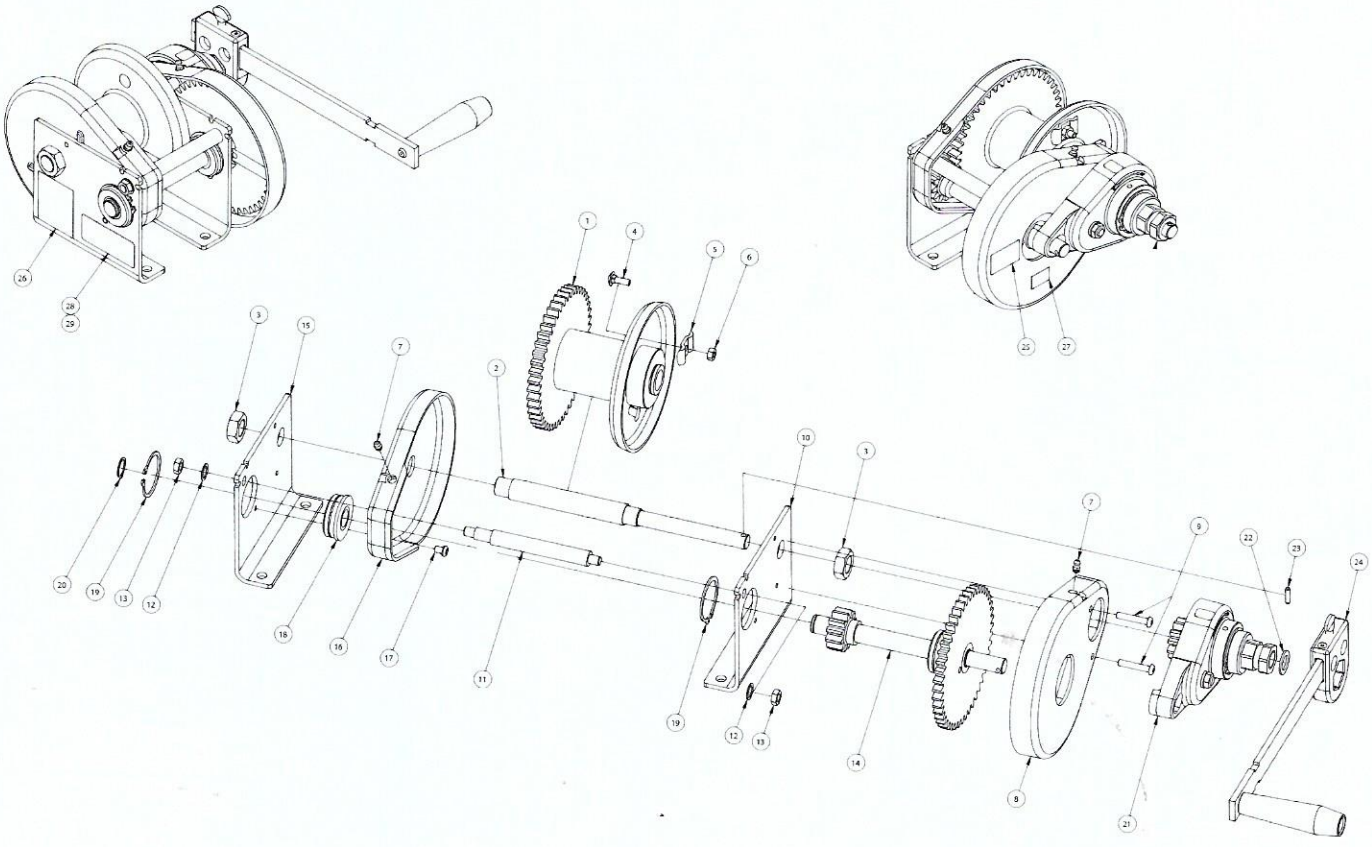
Spur Gear Hand Winch — Model M4412

ITEM	PART NUMBER	QTY	DESCRIPTION	MASS
1	A25689	1	DRUM ASSEMBLY	8.85
2	A25708	1	SHAFT DRUM M4412PB X 3.00 LG DRUM NORLS	1.60
3	A3886	2	NUT HEX JAM .750-16NF SST 18-8	.09
4	A2846	1	CARIGBOLT .250-20NC X .750 SST	.02
5	A5922	1	TIE DOWN	.04
6	A3331	1	NUT HEX .250-20NC SST 18-8	.01
7	A25661	1	ZERK .25-28 UNF SST	.01
8	A25508	1	COUNTER SHAFT GEAR GUARD	.30
9	A25557	2	CAPSCR BTNHD .250-20NC X 1.375 SST 18-8	.02
10	A25614	1	RIGHT SIDE FRAME PLATE	1.91
11	A25695	1	FRAME ROD M4412,M4412PB	.69
12	A3939	2	WASHER SPLK INT .375 X .692 X .035 SST	.00
13	A3330	2	NUT HEX JAM .375-16NC SST 18-8	.01
14	A25692	1	COUNTER SHAFT ASSEMBLY M4412	4.87
15	A25615	1	LEFT SIDE FRAME PLATE	1.91
16	A25559	1	DRUM GEAR GUARD	.24
17	A14146	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
18	A25620	1	BEARING HOLDER ASSEMBLY	.36
19	A3883	2	RING RET EXT 1.750 SST	.01
20	A4408	1	RING RET EXT .750 SST	.00
21	A25680	1	RATCHET PAWL	.08
22	B1145	1	RATCHET SPRING	.01
23	A25704	1	HANDLE ASSEMBLY	1.66
24	A4665	1	PIN COTTER HAIR	.01
25	A4290	1	PIN SLOTTED SPRING .250 X .750 SST	.01
26	10477	1	LABEL DATE OF MANUFACTURE	.00
27	A2175	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
28	A1978	1	LABEL WARNING	.00
29	A25771	1	LABEL MODEL/CAPACITY M4412	.00



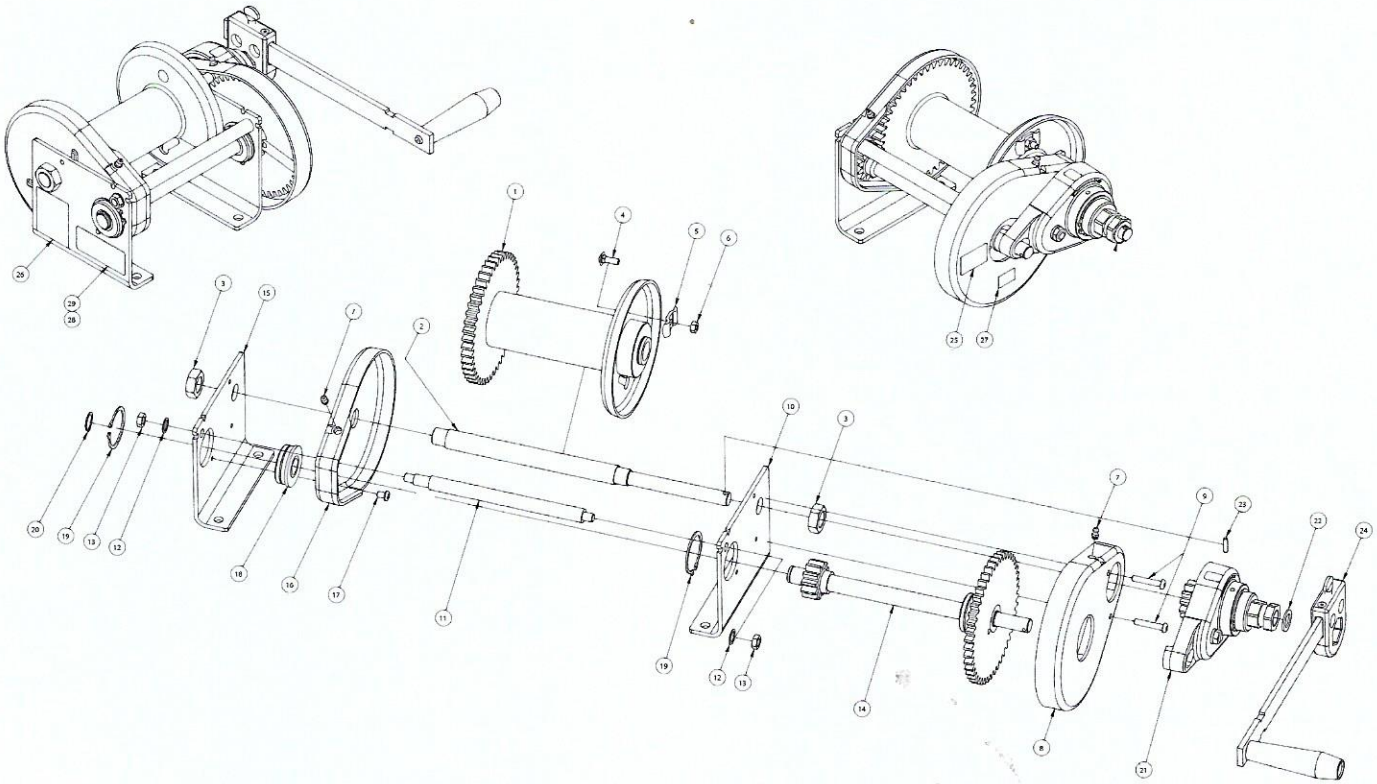
Spur Gear Hand Winch Model M4312PB

ITEM	PART NUMBER	QTY (4312PB)	QTY (4312PB-K)	DESCRIPTION	MASS
1	A25616	1	1	DRUM ASSEMBLY	7.34
2	A25613	1	1	SHAFT DRUM M4311PB X 3.00 LG DRUM NORLS	1.36
3	A3886	2	2	NUT HEX JAM .750-16NF SST 18-8	.09
4	A2846	1	1	CARIGBOLT .250-20NC X .750 SST	.02
5	A5922	1	1	TIE DOWN	.04
6	A3331	1	1	NUT HEX .250-20NC SST 18-8	.01
7	A25661	2	2	ZERK .25-28 UNF SST	.01
8	A25508	1	1	COUNTER SHAFT GEAR GUARD	.30
9	A25557	2	2	CAPSCR BTNHD .250-20NC X 1.375 SST 18-8	.02
10	A25614	1	1	RIGHT SIDE FRAME PLATE	1.91
11	A25627	1	1	FRAME ROD M4312, M4312PB	.43
12	A3939	2	2	WASHER SPLK INT .375 X .692 X .035 SST	.00
13	A3330	2	2	NUT HEX JAM .375-16NC SST 18-8	.01
14	A25626	1	1	COUNTER SHAFT ASSEMBLY	4.49
15	A25615	1	1	LEFT SIDE FRAME PLATE	1.91
16	A25507	1	1	DRUM GEAR GUARD	.26
17	A14146	1	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
18	A25620	1	1	BEARING HOLDER ASSEMBLY	.36
19	A3883	2	2	RING RET EXT 1.750 SST	.01
20	A4408	1	1	RING RET EXT .750 SST	.00
21	A25610	1	1	BRAKE DISC ASSEMBLY	4.00
22	A5059	1	1	MACHBUSH .625 X 1.000 X 14GA SST	.01
23	A4282	1	1	PIN SLOTTED SPRING .187 X .750 SST	.00
24	A25604	1	1	HANDLE MEDALLION ASSEMBLY	2.32
25	A2175	1	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
26	A1978	1	1	LABEL WARNING	.00
27	10477	1	1	LABEL DATE OF MANUFACTURE	.00
28	A25766	1	0	LABEL MODEL/CAPACITY CE M4312PB	.00
29	A25767	0	1	LABEL MODEL/CAPACITY CE M4312PB-K	.00



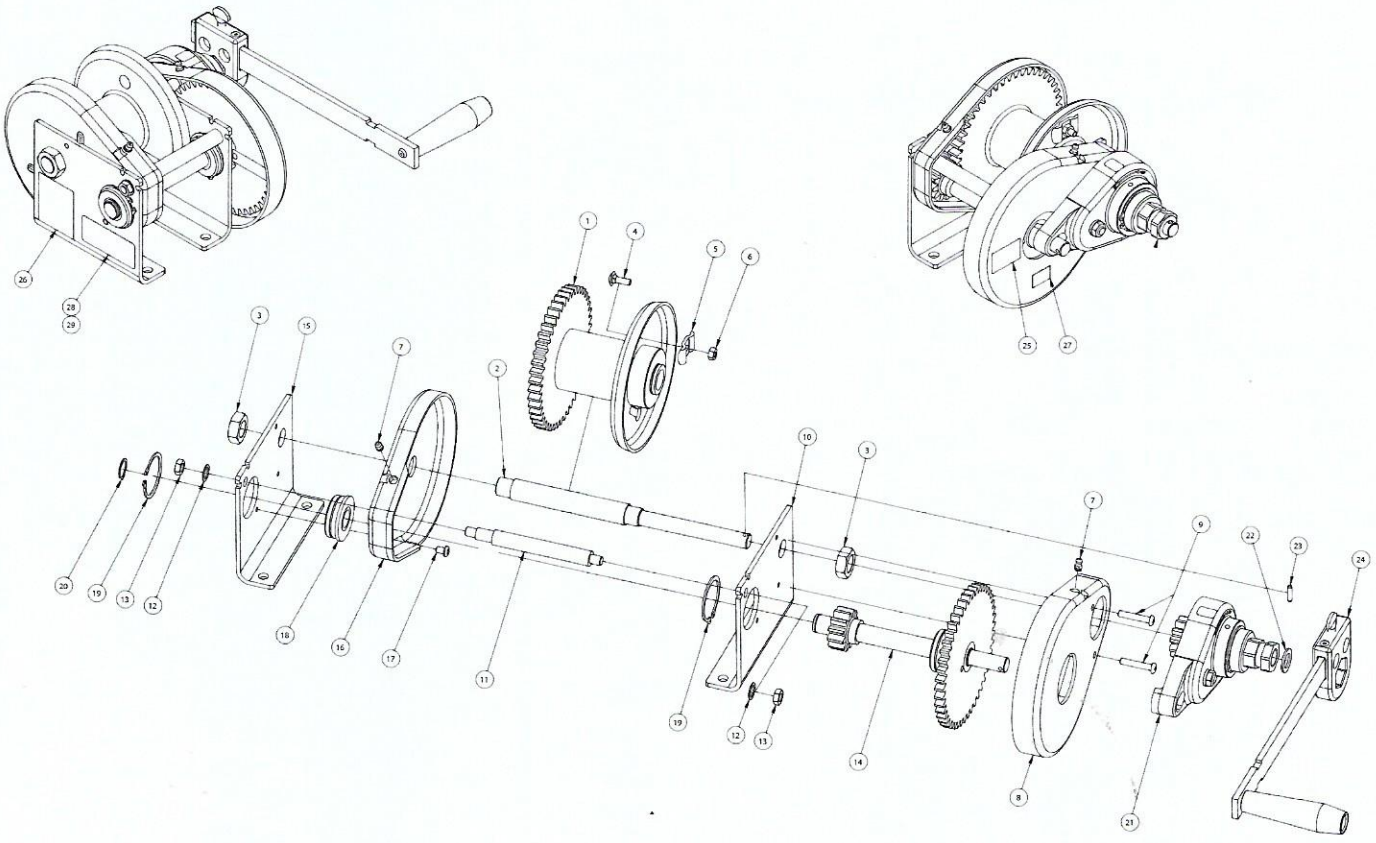
Spur Gear Hand Winch — Model M4412PB

ITEM	PART NUMBER	QTY (4412PB)	QTY (4412PB-K)	DESCRIPTION	MASS
1	A25689	1	1	DRUM ASSEMBLY	8.85
2	A25691	1	1	SHAFT DRUM M4311PB X 3.00 LG DRUM NORLS	1.86
3	A3886	2	2	NUT HEX JAM .750-16NF SST 18-8	.09
4	A2846	1	1	CARIGBOLT .250-20NC X .750 SST	.02
5	A5922	1	1	TIE DOWN	.04
6	A3331	1	1	NUT HEX .250-20NC SST 18-8	.01
7	A25661	2	2	ZERK .25-28 UNF SST	.01
8	A25508	1	1	COUNTER SHAFT GEAR GUARD	.30
9	A25557	2	2	CAPSCR BTNHD .250-20NC X 1.375 SST 18-8	.02
10	A25614	1	1	RIGHT SIDE FRAME PLATE	1.91
11	A25695	1	1	FRAME ROD M4412,M4412PB	.69
12	A3939	2	2	WASHER SPLK INT .375 X .692 X .035 SST	.00
13	A3330	2	2	NUT HEX JAM .375-16NC SST 18-8	.01
14	A25692	1	1	COUNTER SHAFT ASSEMBLY M4412	4.87
15	A25615	1	1	LEFT SIDE FRAME PLATE	1.91
16	A25507	1	1	DRUM GEAR GUARD	.26
17	A14146	1	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
18	A25620	1	1	BEARING HOLDER ASSEMBLY	.36
19	A3883	2	2	RING RET EXT 1.750 SST	.01
20	A4408	1	1	RING RET EXT .750 SST	.00
21	A25610	1	1	BRAKE DISC ASSEMBLY	4.00
22	A5059	1	1	MACHBUSH .625 X 1.000 X 14GA SST	.01
23	A4282	1	1	PIN SLOTTED SPRING .187 X .750 SST	.00
24	A25604	1	1	HANDLE MEDALLION ASSEMBLY	2.32
25	A2175	1	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
26	A1978	1	1	LABEL WARNING	.00
27	10477	1	1	LABEL DATE OF MANUFACTURE	.00
28	A25770	0	1	LABEL MODEL/CAPACITY CE M4412PB-K	.00
29	A25769	1	0	LABEL MODEL/CAPACITY CE M4412PB	.00



Spur Gear Hand Winch Model M4312PBSS

ITEM	PART NUMBER	QTY (4312PBSS)	QTY (4312PBSS-K)	DESCRIPTION	MASS
1	A25575	1	1	DRUM ASSEMBLY	7.50
2	B2474	1	1	SHAFT DRUM M4311PBSS	1.41
3	A3886	2	2	NUT HEX JAM .750-16NF SST 18-8 SST	.09
4	A2846	1	1	CARIGBOLT .250-20NC X .750 SST	.02
5	A5922	1	1	TIE DOWN	.04
6	A3331	1	1	NUT HEX .250-20NC SST 18-8 SST	.01
7	A25661	2	2	ZERK .25-28 UNF SST	.01
8	A25508	1	1	COUNTER SHAFT GEAR GUARD	.30
9	A25557	2	2	CAPSCR BTNHD .250-20NC X 1.375 SST	.02
10	A25664	1	1	RIGHT SIDE FRAME PLATE	1.95
11	B2476	1	1	FRAME ROD M4312, M4312PB	.37
12	A3939	2	2	WASHER SPLK INT .375 X .692 X .035 SST	.00
13	A3330	2	2	NUT HEX JAM .375-16NC 18-8 SST	.01
14	B2472	1	1	COUNTER SHAFT ASSEMBLY	4.56
15	A25615	1	1	LEFT SIDE FRAME PLATE	1.95
16	A25507	1	1	DRUM GEAR GUARD	.26
17	A14146	1	1	CAPSCR BTNHD .250-20NC X .375 SST	.01
18	A4599	1	1	BEARING HOLDER ASSEMBLY	.37
19	A3883	2	2	RING RET EXT 1.750 SST	.01
20	A4408	1	1	RING RET EXT .750 SST	.00
21	A27223	1	1	DISC BRAKE ASSEMBLY	4.28
22	A5059	1	1	MACHBUSH .625 X 1.000 X 14GA SST	.01
23	A4282	1	1	PIN SLOTTED SPRING .187 X .750 SST	.00
24	B4217	1	1	HANDLE MEDALLION ASSEMBLY	2.42
25	10477	1	1	LABEL DATE OF MANUFACTURE	.00
26	A2175	1	1	LABEL CAUTION THIS UNIT IS NOT FACTORY LUBRICATED	.00
27	A1978	1	1	LABEL WARNING	.00
28	A25774	1	0	LABEL MODEL/CAPACITY CE M4312PBSS	.00
29	A25775	0	1	LABEL MODEL/CAPACITY CE M4312PBSS-K	.00



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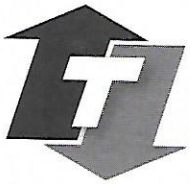

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